TM Forum Specification

Private Optimized Binding

TMF759

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Introduction

The following document is the user guide of the REST API for Any management. It includes the model definition as well as all available operations.

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Sample Use Cases

Private Optimized Binding Use Case for security company

What is the business context?

Network-as-a-Service (NaaS) platforms allow operators to expose telco capabilities to 3rd party service providers in a programmatic manner, through Open Application Programming Interfaces (APIs). These Open APIs pave the way for transforming telco networks into programmable service platforms, enabling the integration of the network with 3rd party applications, with frictionless interactions between them. This is a win-win situation for the two stakeholders involved. For CSPs, this represents a business opportunity to generate new revenue streams, and one of the ways to better monetize investment in 5G, access, and edge computing. For 3rd parties, it allows them to flee from traditional over-the-top, best-effort service delivery approaches, tapping now into offered capabilities to provide enhanced user experiences and contribute to the digital ecosystem with new services.

What is the purpose?

The existing process for purchasing network services from Communications Service Providers today is not aligned with customer expectations for a digital-first, on-demand experience. CSPs need to decompose their network assets and make network services available as API-enabled products. The network needs to be elastic to allows for bandwidth, scaling and healing. For example, a CSP's business customer who provides security guards and services for special events at stadiums or public venues needs to have access to high-quality video on demand. Developer of the cloud-based video security app can use NaaS APIs to get performance data from the network and set a policy to configure an automatic upgrade of service, or upgrade manually at the tap of a button.

Where is the value?

Creation of a new Tmforum NaaS API that will ensure that an application, deployed on the Cloud by a 3rd party finds a path to the most performant broadband connection at the right time. The fastest broadband connection is determined based on where the device/customer is located and the most performant and available local network connection. By identifying network APIs that customers really want, and standardizing them across the industry, it will open up new lines of business for CSPs, Hyperscalers and App producers by improving the experience of purchasing network services, drive innovation in the industry, and increase the total addressable market for these services through the introduction of value-added products.

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Use Case: Special Events Security Company



I need to stream 4K video

- ✓ A CSP's business customer provides security guards and services for special events at stadiums, parks, racetracks, fairgrounds, etc.
- ✓ Brings their own wireless cameras and communications equipment to ensure coverage across outdoor areas like parking lots.
- ✓ Uploads live footage to a cloud-based app for analyzing video feeds and streaming to guards' mobile devices → occasional heavy usage.
- ✓ Needs priority service at congested events, but on an irregular schedule.
- ✓ Standard wireless service doesn't consistently have high quality of service.



Monitor

TMF 628



- ✓ Developer of the cloud-based video security app can get performance data from the network, even if a camera or device has a poor connection.
- ✓ The Insights API quantifies the performance through the CSP's network.
- ✓ To ensure timely availability of high-quality video, the security company can set a policy to configure an automatic upgrade of service, or upgrade manually at the tap of a button.
- ✓ Insights API confirms better performance after the temporary upgrade.

PLP Scope

- ✓ The Device UPF assignment is optimized for performance in the CSP network (TMF 759)
- ✓ The UPF connection to the Enterprise App is established against its closest region/edge presence
- ✓ Establishes end to end Data Path
- ✓ Users are able to access the Enterprise App in leveraging Private Network Port

TMF 759

- ✓ Metering API senses the user's session has neared the usage window and prompts for extension or termination

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Support of polymorphism and extension patterns

Support of polymorphic collections and types and schema based extension is provided by means of a list of generic meta-attributes that we describe below. Polymorphism in collections occurs when entities inherit from base entities, for instance a BillingAccount and SettlementAccount inheriting properties from the abstract Account entity.

Generic support of polymorphism and pattern extensions is described in the TMF API Guidelines, Part 2 (TMF630).

The @type attribute provides a way to represent the actual class type of an entity. For example, within a list of Account instances some may be instances of BillingAccount where other could be instances of SettlementAccount. The @type gives this information. All resources and sub-resources of this API have a @type attributes that can be provided when this is useful.

The @referredType can be used within reference entities (like for instance an AccountRef object) to explicitly denote the actual entity type of the referred class. Notice that in reference entities the @type, when used, denotes the class type of the reference itself, such as BillingAccountRef or SettlementAccountRef, and not the class type of the referred object. However since reference classes are rarely sub-classed, @type is generally not useful in reference objects.

The @schemaLocation property can be used in resources to allow specifying user-defined properties of an Entity or to specify the expected characteristics of an entity.

The @baseType attribute gives a way to provide explicitly the base of class of a given resource that has been extended.

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RESOURCE MODEL

Managed Entity and Task Resource Models

UserEquipment resource

UserEquipment is a type of physical resource that describes any type of end user equipment in TMF759 binding.

Resource model

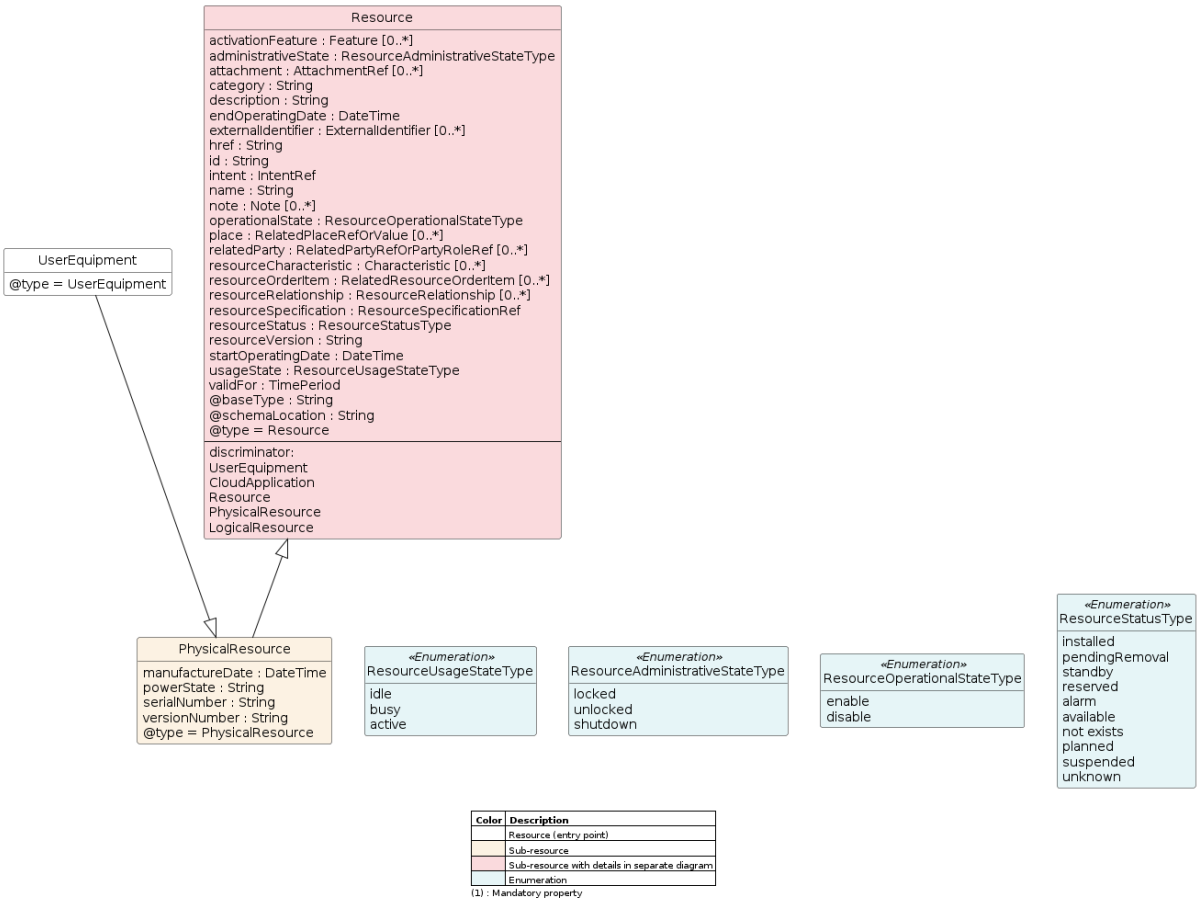


Figure 1 - UserEquipment

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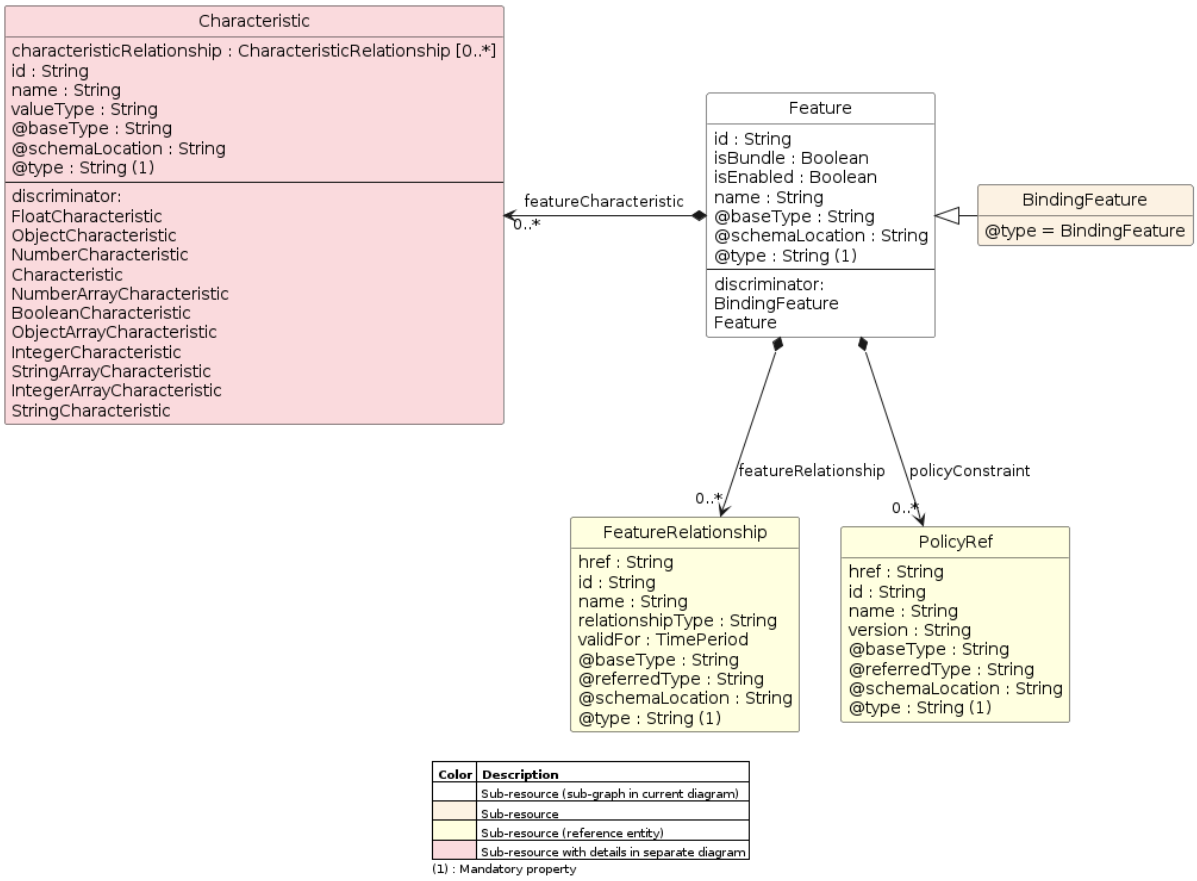


Figure 2 - Feature

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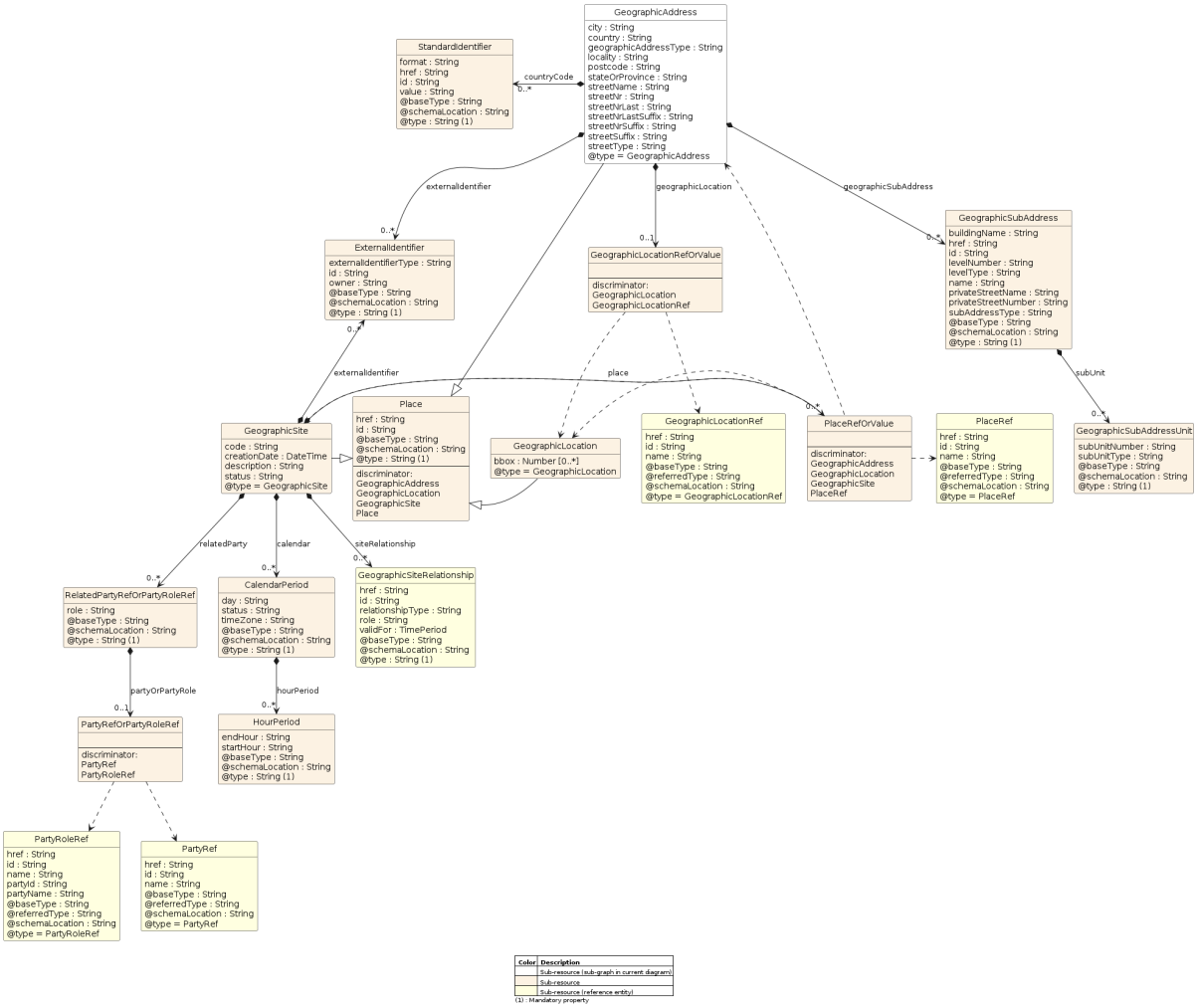


Figure 3 - GeographicAddress

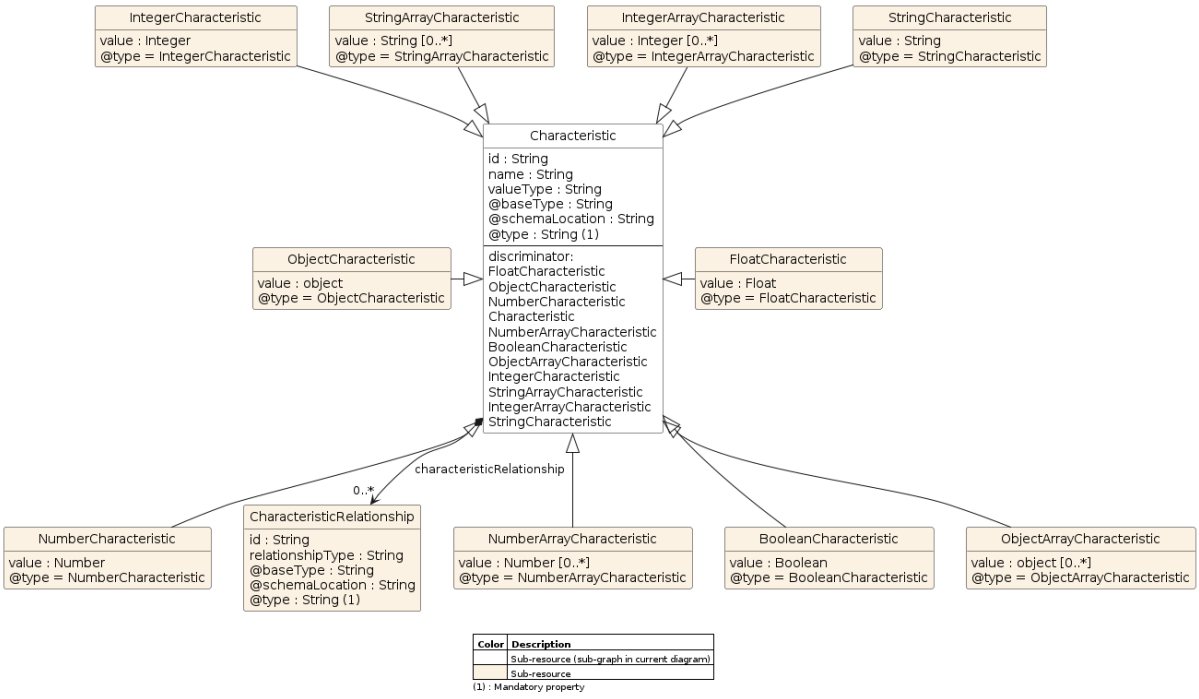


Figure 4 - Characteristic

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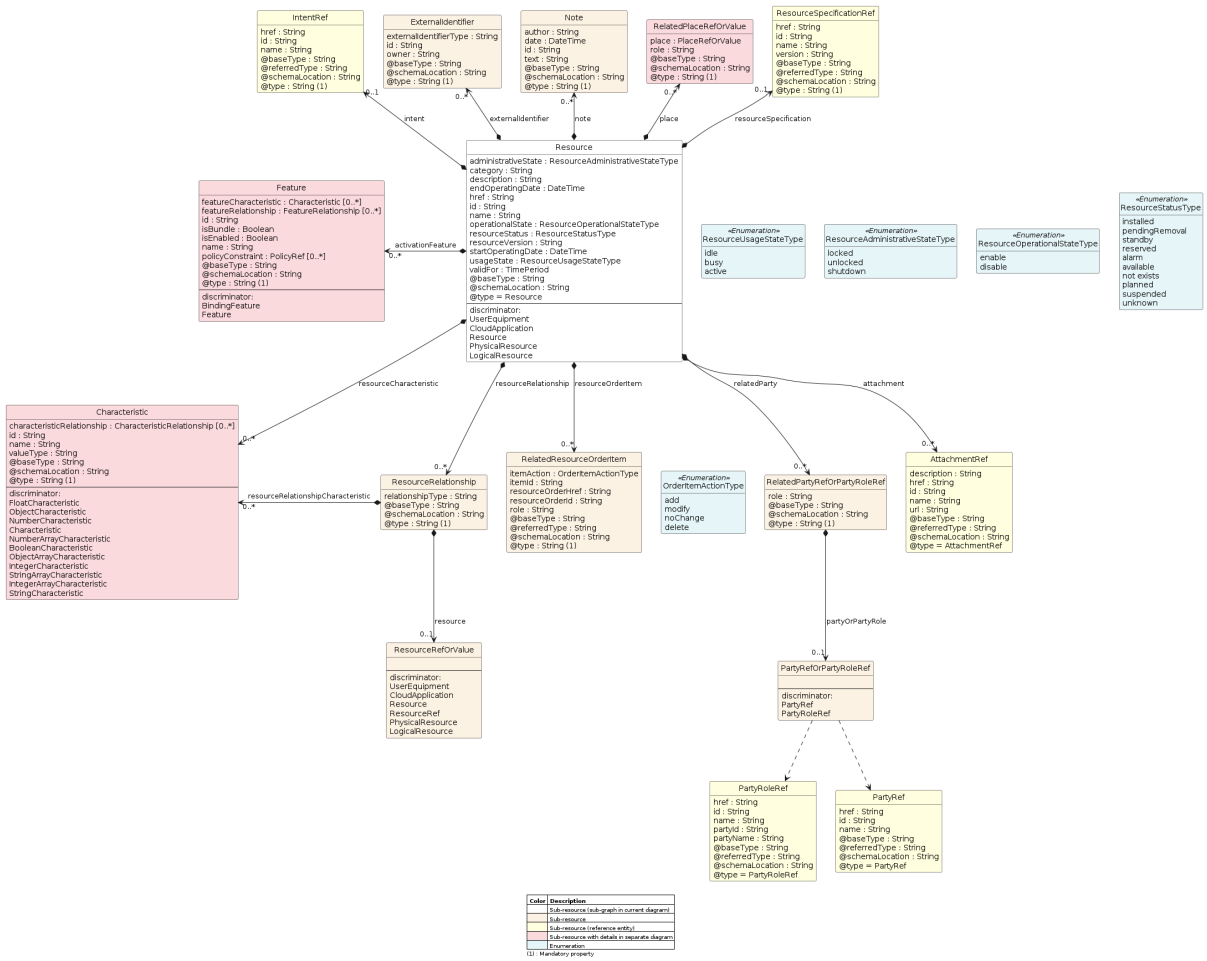


Figure 5 - Resource

Field descriptions

UserEquipment fields

administrativeState	A ResourceAdministrativeStateType. ResourceAdministrativeStateType enumerations.
operationalState	A ResourceOperationalStateType. ResourceOperationalStateType enumerations.
resourceStatus	A ResourceStatusType. ResourceStatusType enumerations.
usageState	A ResourceUsageStateType. ResourceUsageStateType enumerations.

AttachmentRef sub-resource fields

description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.

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url	A String. Link to the attachment media/content.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BooleanCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Boolean. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CalendarPeriod sub-resource fields

day	A String. Day where the calendar status applies (e.g.: monday, mon-to-fri, weekdays, weekend, all week, …).
hourPeriod	A HourPeriod.
status	A String. Indication of the availability of the caledar period (e.g.: available, booked, etc.).
timeZone	A String. Indication of the timezone applicable to the calendar information (e.g.: Paris, GMT+1).
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Characteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.

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valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CharacteristicRelationship sub-resource fields

id	A String. Unique identifier of the characteristic.
relationshipType	A String. The type of relationship.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ExternalIdentifier sub-resource fields

externalIdentifierType	A String. Type of the identification, typically would be the type of the entity within the external system.
id	A String. Identification of the entity within the external system.
owner	A String. Name of the external system that owns the entity.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Feature sub-resource fields

featureCharacteristic	A Characteristic. This is a list of Characteristics for a particular feature.
featureRelationship	A FeatureRelationship.
id	A String. Unique identifier.
isBundle	A Boolean. True if this is a feature group. Default is false.
isEnabled	A Boolean. True if this feature is enabled. Default is true.
name	A String. This is the name for the feature.
policyConstraint	A PolicyRef. Reference to mangage a list of feature specification policy constraints.
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FeatureRelationship sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
relationshipType	A String. This is the type of the feature relationship.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FloatCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Float. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicAddress sub-resource fields

city	A String. City that the address is in.
country	A String. Country that the address is in.

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countryCode	A StandardIdentifier. Country codes are short alphabetic or numeric geographical codes (geocodes) developed to represent countries and dependent areas, for use in data processing and communications. Several different systems have been developed to do this. The term country code frequently refers to ISO 3166-1 alpha-2 or international dialing codes, the E.164 country calling codes.
externalIdentifier	An ExternalIdentifier.
geographicAddressType	A String. Classification of the address, e.g., residential, industrial.
geographicLocation	A GeographicLocationRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the GeographicLocation entity and not the GeographicLocationRefOrValue class itself.
geographicSubAddress	A GeographicSubAddress.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
locality	A String. An area of defined or undefined boundaries within a local authority or other legislatively defined area, usually rural or semi rural in nature. [ANZLIC-STREET], or a suburb, a bounded locality within a city, town or shire principally of urban character [ANZLICSTREET].
postcode	A String. Descriptor for a postal delivery area, used to speed and simplify the delivery of mail (also know as zipcode).
stateOrProvince	A String. The State or Province that the address is in.
streetName	A String. Name of the street or other street type.
streetNr	A String. Number identifying a specific property on a public street. It may be combined with streetNrLast for ranged addresses.
streetNrLast	A String. Last number in a range of street numbers allocated to a property.
streetNrLastSuffix	A String. Last street number suffix for a ranged address.
streetNrSuffix	A String. The first street number suffix.
streetSuffix	A String. A modifier denoting a relative direction.
streetType	A String. Alley, avenue, boulevard, brae, crescent, drive, highway, lane, terrace, parade, place, tarn, way, wharf.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicLocation sub-resource fields

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bbox	A Number. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicLocationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicSite sub-resource fields

calendar	A CalendarPeriod.
code	A String. A code that may be used for some addressing schemes eg: [ANSI T1.253-1999].
creationDate	A DateTime. Date and time when the GeographicSite was created.
description	A String. Text describing additional information regarding the site.
externalIdentifier	An ExternalIdentifier.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
place	A PlaceRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Place entity and not the PlaceRefOrValue class itself.
relatedParty	A RelatedPartyRefOrPartyRoleRef.
siteRelationship	A GeographicSiteRelationship.
status	A String. The condition of the GeographicSite, such as planned, underConstruction, cancelled, active, inactive, former.

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@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicSiteRelationship sub-resource fields

href	A String. Reference of the related geographic site.
id	A String. Unique identifier of the related site entity within the server.
relationshipType	A String. Type of relationship.
role	A String. Role of the related site in the relationship.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicSubAddress sub-resource fields

buildingName	A String. Allows for buildings that have well-known names.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
levelNumber	A String. Used where a level type may be repeated e.g. BASEMENT 1, BASEMENT 2.
levelType	A String. Describes level types within a building.
name	A String. Name of the subAddress to identify it with a meaningful identification.
privateStreetName	A String. Private streets internal to a property (e.g. a university) may have internal names that are not recorded by the land title office.
privateStreetNumber	A String. Private streets numbers internal to a private street.
subAddressType	A String. Type of subAddress : it can be a subunit or a private street.
subUnit	A GeographicSubAddressUnit. Representation of a SubUnit. It is used for describing subunit within a subAddress e.g. BERTH, FLAT, PIER, SUITE, SHOP, TOWER, UNIT, WHARF.
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicSubAddressUnit sub-resource fields

subUnitNumber	A String. The discriminator used for the subunit, often just a simple number but may also be a range.
subUnitType	A String. The type of subunit e.g.BERTH, FLAT, PIER, SUITE, SHOP, TOWER, UNIT, WHARF, RACK.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

HourPeriod sub-resource fields

endHour	A String. The time when the status ends applying.
startHour	A String. The time when the status starts applying.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An Integer.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship.

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id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An Integer. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntentRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Note sub-resource fields

author	A String. Author of the note.
date	A DateTime. Date of the note.
id	A String. Identifier of the note within its containing entity.
text	A String. Text of the note.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Number.

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valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Number. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An object.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An object. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.

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@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRoleRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
partyId	A String. The identifier of the engaged party that is linked to the PartyRole object.
partyName	A String. The name of the engaged party that is linked to the PartyRole object.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PhysicalResource sub-resource fields

activationFeature	A Feature. Configuration features.
administrativeState	A ResourceAdministrativeStateType. ResourceAdministrativeStateType enumerations.
attachment	An AttachmentRef.

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category	A String. Category of the concrete resource. e.g Gold, Silver for MSISDN concrete resource.
description	A String. Free-text description of the resource.
endOperatingDate	A DateTime. A date time(DateTime). The date till the resource is operating.
externalIdentifier	An ExternalIdentifier. An identification of this resource that is owned by or originates in a software system different from the current system. The structure identifies the system itself, the nature of the resource within the system and the unique ID of the resource within the system. It is anticipated that multiple external IDs can be held for a single resource, e.g. if the resource passed through multiple systems on the way to the current system.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intent	An IntentRef. Intent reference, for when Intent is used by other entities.
manufactureDate	A DateTime. This is a string attribute that defines the date of manufacture of this item in the fixed format "dd/mm/yyyy". This is an optional attribute.
name	A String. The name of the resource.
note	A Note.
operationalState	A ResourceOperationalStateType. ResourceOperationalStateType enumerations.
place	A RelatedPlaceRefOrValue.
powerState	A String. This defines the current power status of the hardware item. Values include: 0: Unknown 1: Not Applicable 2: No Power Applied 3: Full Power Applied 4: Power Save - Normal 5: Power Save - Degraded 6: Power Save - Standby 7: Power Save - Critical 8: Power Save - Low Power Mode 9: Power Save - Unknown 10: Power Cycle 11: Power Warning 12: Power Off.
relatedParty	A RelatedPartyRefOrPartyRoleRef.
resourceCharacteristic	A Characteristic.
resourceOrderItem	A RelatedResourceOrderItem. A list of resource order items related to this resource.
resourceRelationship	A ResourceRelationship.

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resourceSpecification	A ResourceSpecificationRef.
resourceStatus	A ResourceStatusType. ResourceStatusType enumerations.
resourceVersion	A String. A field that identifies the specific version of an instance of a resource.
serialNumber	A String. This is a string that represents a manufacturer-allocated number used to identify different instances of the same hardware item. The ModelNumber and PartNumber attributes are used to identify different types of hardware items. This is a REQUIRED attribute.
startOperatingDate	A DateTime. A date time(DateTime). The date from which the resource is operating.
usageState	A ResourceUsageStateType. ResourceUsageStateType enumerations.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
versionNumber	A String. This is a string that identifies the version of this physical resource. This is an optional attribute.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Place sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PlaceRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String. When sub-classing, this defines the sub-class Extensible name.

PolicyRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyRefOrPartyRoleRef sub-resource fields

partyOrPartyRole	A PartyRefOrPartyRoleRef.
role	A String. Role played by the related party or party role in the context of the specific entity it is linked to. Such as 'initiator', 'customer', 'salesAgent', 'user'.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPlaceRefOrValue sub-resource fields

place	A PlaceRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Place entity and not the PlaceRefOrValue class itself.
role	A String.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedResourceOrderItem sub-resource fields

itemAction	An OrderItemActionType. Action to be performed on the product.

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itemId	A String. Identifier of the order item where the resource was managed.
resourceOrderHref	A String. Reference of the related entity.
resourceOrderId	A String. Unique identifier of a related entity.
role	A String. Role of the resource order item for this resource.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Resource sub-resource fields

activationFeature	A Feature. Configuration features.
administrativeState	A ResourceAdministrativeStateType. ResourceAdministrativeStateType enumerations.
attachment	An AttachmentRef.
category	A String. Category of the concrete resource. e.g Gold, Silver for MSISDN concrete resource.
description	A String. Free-text description of the resource.
endOperatingDate	A DateTime. A date time(DateTime). The date till the resource is operating.
externalIdentifier	An ExternalIdentifier. An identification of this resource that is owned by or originates in a software system different from the current system. The structure identifies the system itself, the nature of the resource within the system and the unique ID of the resource within the system. It is anticipated that multiple external IDs can be held for a single resource, e.g. if the resource passed through multiple systems on the way to the current system.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intent	An IntentRef. Intent reference, for when Intent is used by other entities.
name	A String. The name of the resource.
note	A Note.
operationalState	A ResourceOperationalStateType. ResourceOperationalStateType enumerations.
place	A RelatedPlaceRefOrValue.
relatedParty	A RelatedPartyRefOrPartyRoleRef.
resourceCharacteristic	A Characteristic.

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resourceOrderItem	A RelatedResourceOrderItem. A list of resource order items related to this resource.
resourceRelationship	A ResourceRelationship.
resourceSpecification	A ResourceSpecificationRef.
resourceStatus	A ResourceStatusType. ResourceStatusType enumerations.
resourceVersion	A String. A field that identifies the specific version of an instance of a resource.
startOperatingDate	A DateTime. A date time(DateTime). The date from which the resource is operating.
usageState	A ResourceUsageStateType. ResourceUsageStateType enumerations.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceRelationship sub-resource fields

relationshipType	A String. Type of the resource relationship, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (eg: an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.
resource	A ResourceRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Resource entity and not the ResourceRefOrValue class itself.
resourceRelationshipCharacteristic	A Characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.

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version	A String. Resource Specification version.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StandardIdentifier sub-resource fields

format	A String. Standard/Regulatory definition identifier. e.g., ISO 3166-1.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
value	A String. The value of the resource in the corresponding standard.e.g., a country code value.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A String.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.

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value	A String. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'UserEquipment' resource object.


```
{
    "id": "1114",
    "href": "http://server:port/privateOptimizedBinding/v5/userEquipment/1114",
    "description": "This is an example of a user equipment that we want to bind to a
cloud application.",
    "name": "Security Device XYZ",
    "operationalState": "enable",
    "note": [{
        "text": "something about this resource",
        "@type": "Note"
    }],
    "place": [{
            "role": "HeadQuarter",
            "place": {
                "id": "2b8cea8f-642d-42e0-bdb2-6f47fabb3f07",
                "href": "https://host/tmf-
api/geographicAddressManagement/v4/geographicAddress/2b8cea8f-642d-42e0-bdb2-
6f47fabb3f07",
                "streetNr": "225",
                "streetNrSuffix": "B",
                "streetNrLast": "",
                "streetNrLastSuffix": "",
                "streetName": " Strathmore",
                "streetType": "Terrace",
                "streetSuffix": "",
                "postcode": "5004",
                "locality": "Brighton.",
                "city": "Brighton",
                "stateOrProvince": "SA",
                "country": "Australia",
                "@type": "GeographicAddress"
            },
            "@type": "RelatedPlaceRefOrValue"
    }],
    "activationFeature": [{
            "id": "2",
            "isEnabled": true,
            "name": "randomFeature",
            "featureCharacteristic": [{
                    "name": "testCharecteristic",
                    "value": "testValue",
                    "@type" : "StringCharacteristic"
                }],
```


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```
            "@type" : "Feature"
        }],
    "resourceCharacteristic": [{
        "name": "MSISDN",
        "value": "3345667788643",
        "@type": "StringCharacteristic"
    }],
    "resourceSpecification" : {
        "id" : "11140",
        "href":
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11140",
        "@type" : "ResourceSpecificationRef"
    },
    "@type": "UserEquipment",
    "@schemaLocation": "http://server:port/UserEquipment.schema.json",
    "@baseType": "PhysicalResource"
}
```

We provide below a JSON representation as example of the 'UserEquipment' resource object.

```
{
    "id": "1114",
    "href": "http://server:port/privateOptimizedBinding/v5/userEquipment/1114",
    "description": "This is an example of a user equipment that we want to bind to a
cloud application.",
    "name": "Security Device XYZ",
    "operationalState": "enable",
    "note": [{
            "text": "something about this resource",
            "@type": "Note"
        }],
    "place": [{
        "role": "HeadQuarter",
        "place": {
            "id": "2b8cea8f-642d-42e0-bdb2-6f47fabb3f07",
            "href": "https://host/tmf-
api/geographicAddressManagement/v4/geographicAddress/2b8cea8f-642d-42e0-bdb2-
6f47fabb3f07",
            "streetNr": "225",
            "streetNrSuffix": "B",
            "streetNrLast": "",
            "streetNrLastSuffix": "",
            "streetName": " Strathmore",
            "streetType": "Terrace",
            "streetSuffix": "",
            "postcode": "5004",
            "locality": "Brighton.",
            "city": "Brighton",
            "stateOrProvince": "SA",
            "country": "Australia",
            "@type": "GeographicAddress"
        },
        "@type": "RelatedPlaceRefOrValue"
    }],
    "activationFeature": [{
            "id": "1",
            "isEnabled": true,
            "name": "privateOptimizedBindingFeature",
            "featureCharacteristic": [{
```

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```
                    "name": "privateOptimizedBindingCloudAppHref",
                    "value":
"http://server:port/privateOptimizedBinding/v5/cloudApplication/2287",
                    "@type" : "StringCharacteristic"
                }],
            "@type": "BindingFeature"
        },
        {
            "id": "2",
            "isEnabled": true,
            "name": "randomFeature",
            "featureCharacteristic": [{
                    "name": "testCharecteristic",
                    "value": "testValue",
                    "@type" : "StringCharacteristic"
                }],
            "@type" : "Feature"
        }],
    "resourceCharacteristic": [{
            "name": "MSISDN",
            "value": "336094512412345",
            "@type": "StringCharacteristic"
        }],
    "resourceSpecification" : {
        "id" : "11143",
        "href" :
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11143",
        "@type" : "ResourceSpecificationRef"
    },
    "@type": "UserEquipment",
    "@schemaLocation": "http://server:port/UserEquipment.schema.json",
    "@baseType": "PhysicalResource"
}
```


CloudApplication resource

Cloud Application is a type of resource that describes the cloud application part of a binding in TMF759.

Resource model

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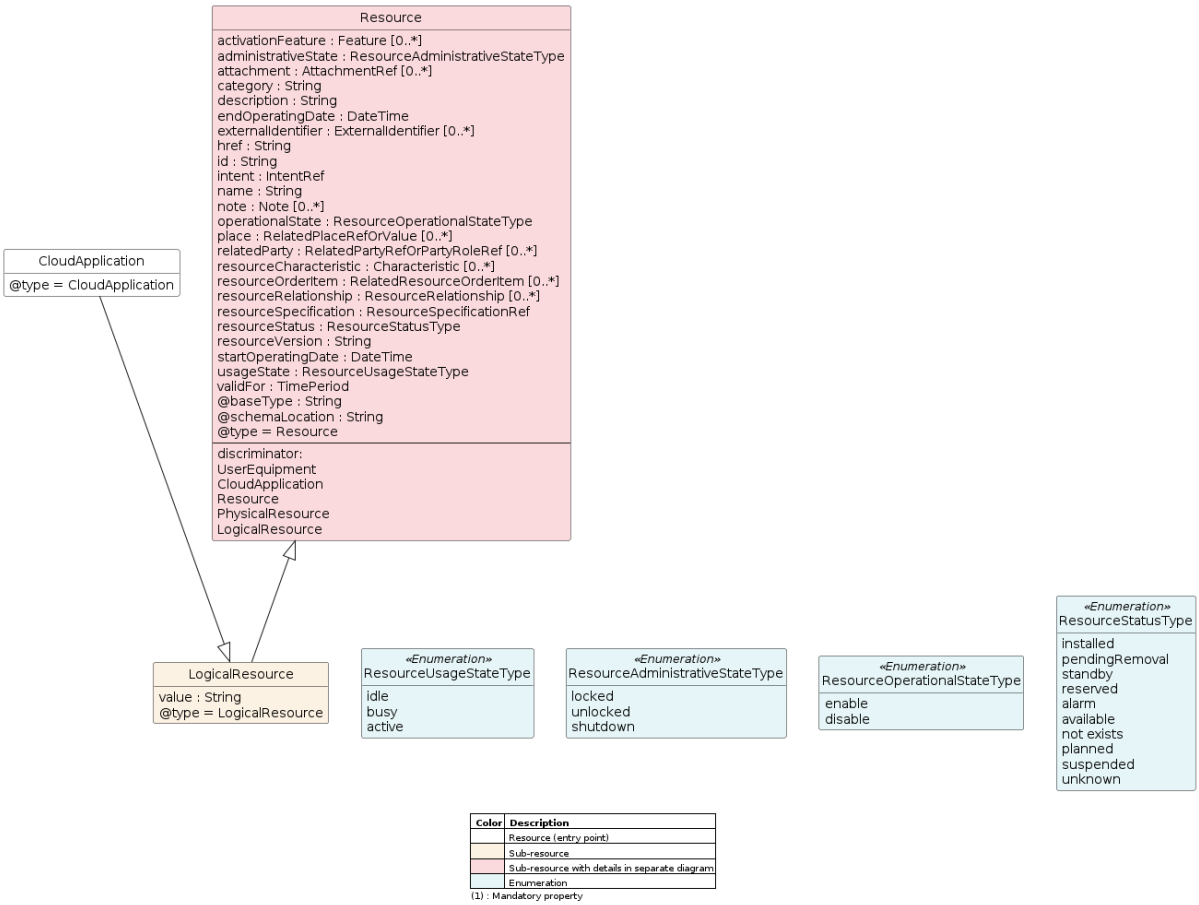


Figure 6 - CloudApplication

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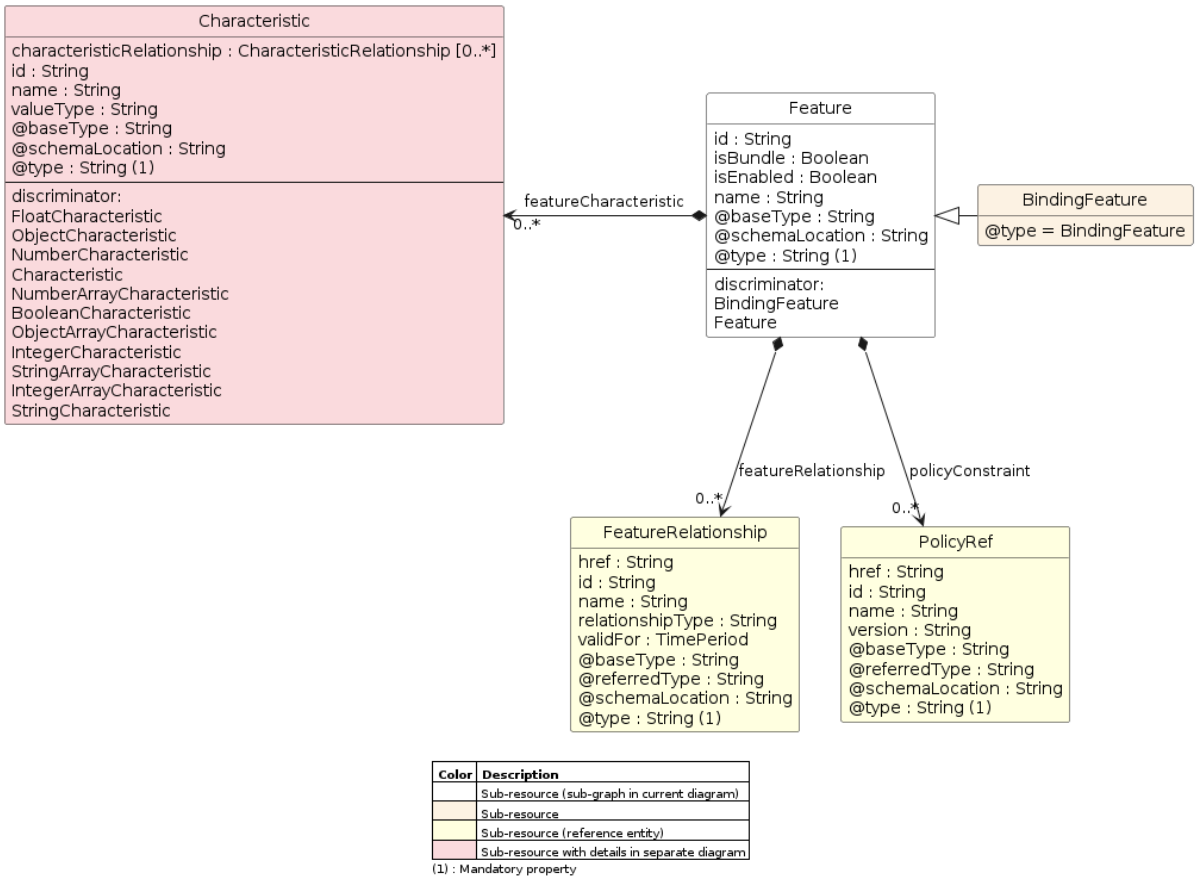


Figure 7 - Feature

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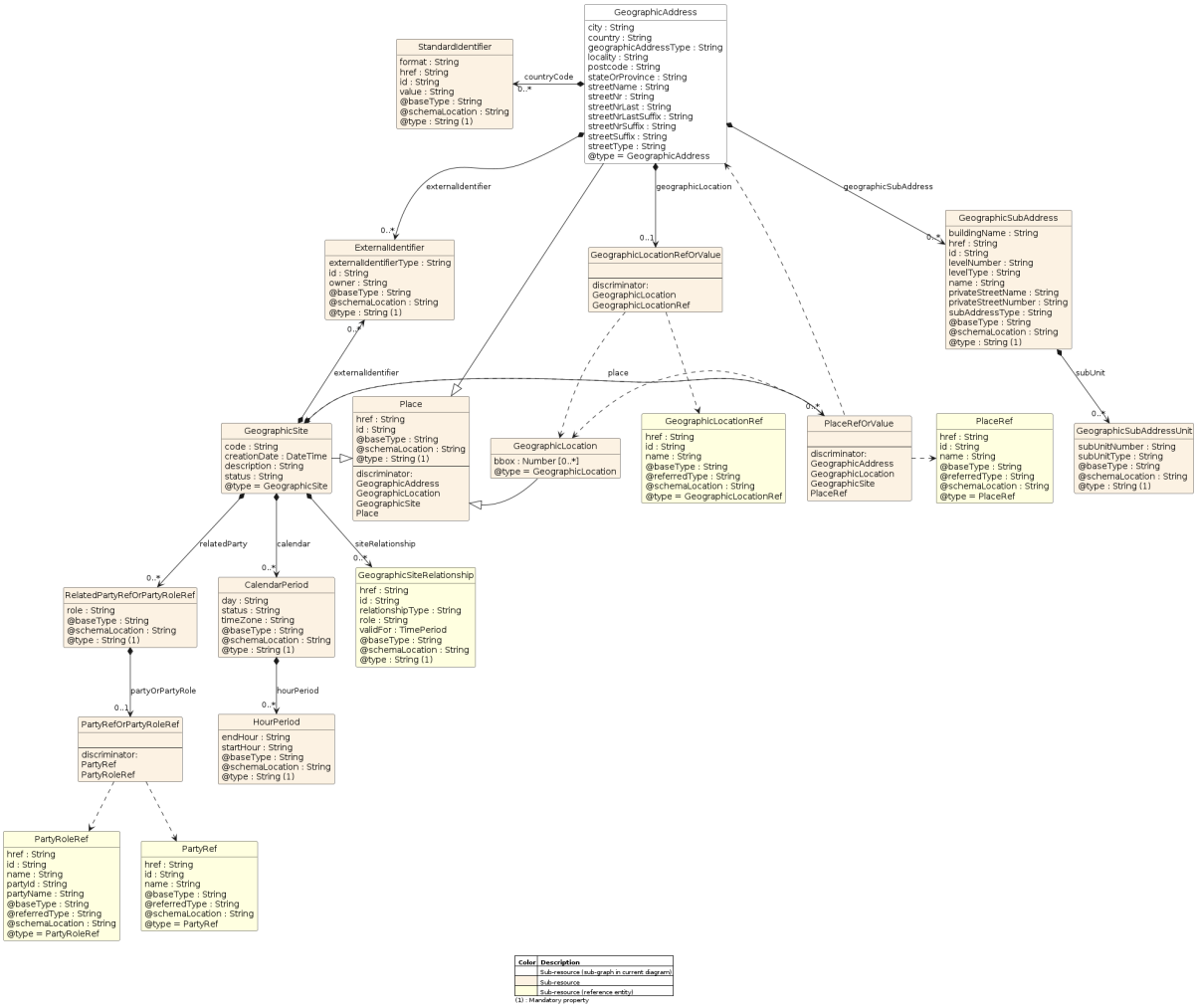


Figure 8 - GeographicAddress

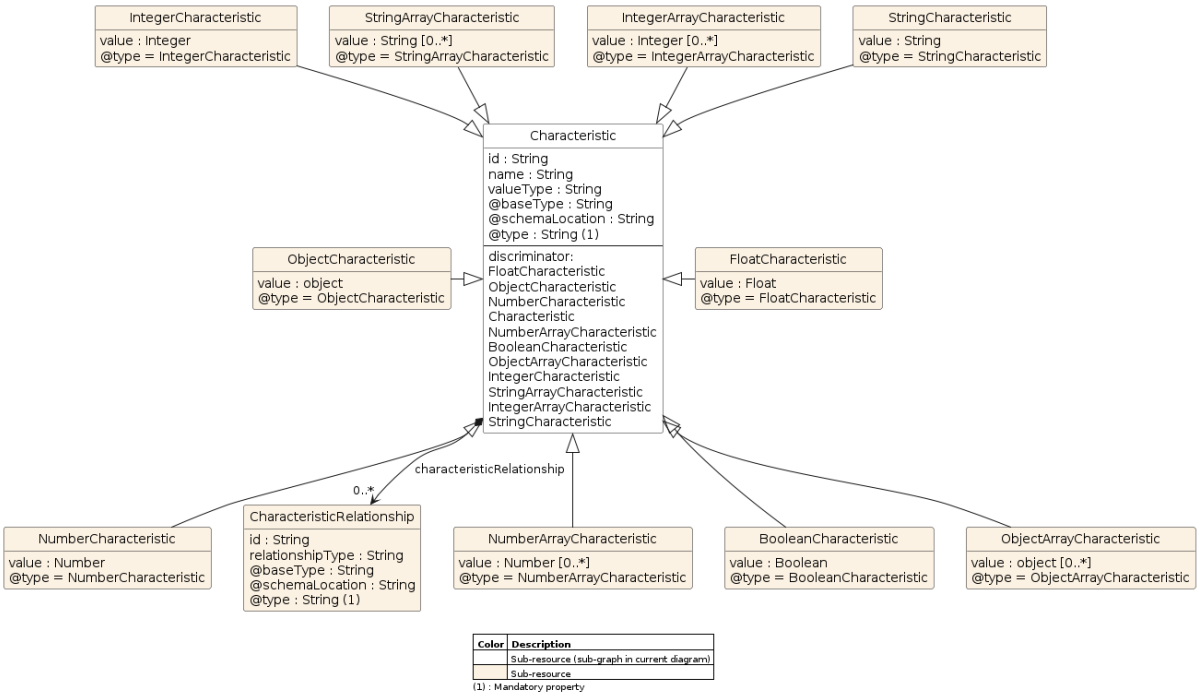


Figure 9 - Characteristic

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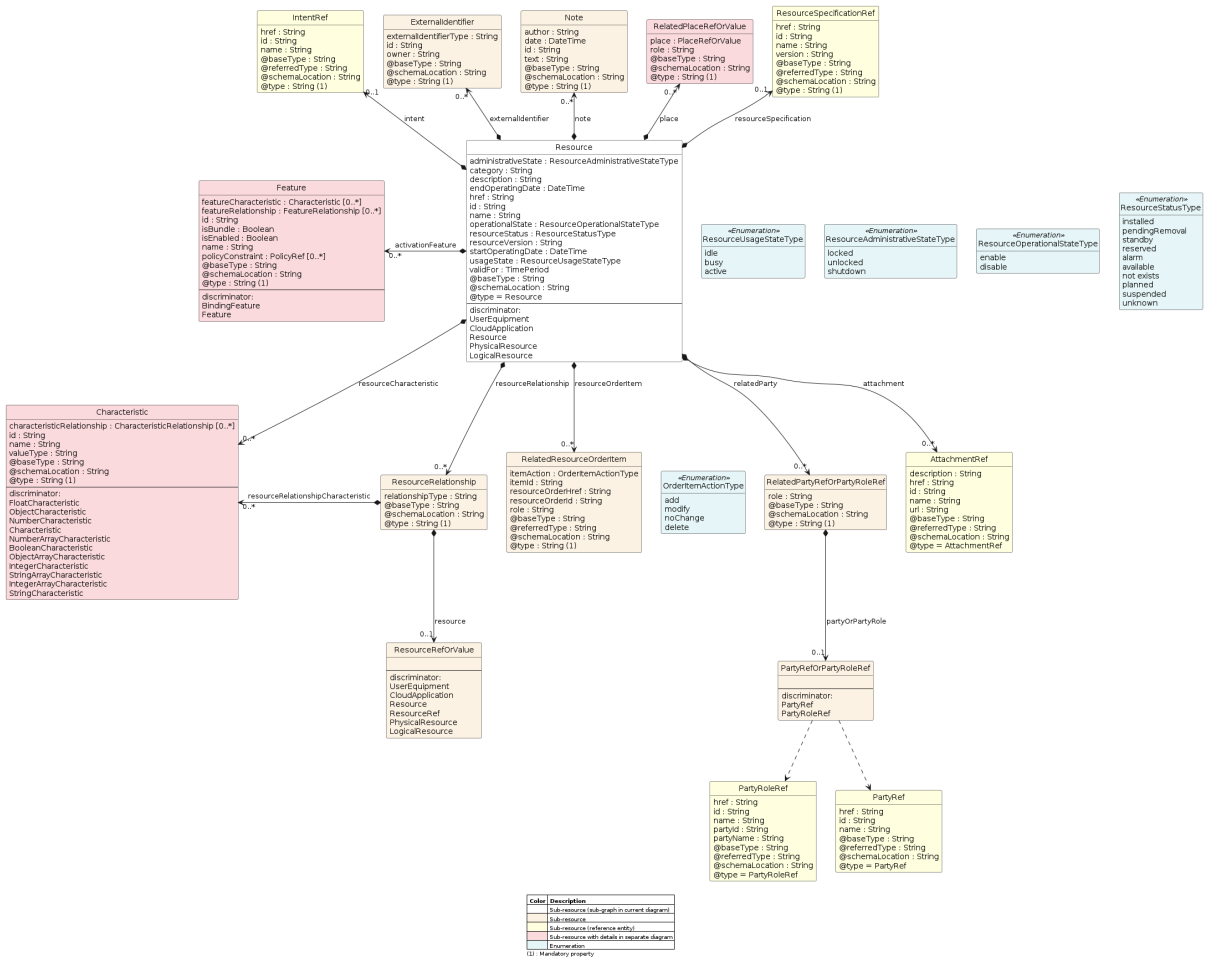


Figure 10 - Resource

Field descriptions

CloudApplication fields

administrativeState	A ResourceAdministrativeStateType. ResourceAdministrativeStateType enumerations.
operationalState	A ResourceOperationalStateType. ResourceOperationalStateType enumerations.
resourceStatus	A ResourceStatusType. ResourceStatusType enumerations.
usageState	A ResourceUsageStateType. ResourceUsageStateType enumerations.

AttachmentRef sub-resource fields

description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.

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url	A String. Link to the attachment media/content.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

BooleanCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Boolean. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CalendarPeriod sub-resource fields

day	A String. Day where the calendar status applies (e.g.: monday, mon-to-fri, weekdays, weekend, all week, …).
hourPeriod	A HourPeriod.
status	A String. Indication of the availability of the caledar period (e.g.: available, booked, etc.).
timeZone	A String. Indication of the timezone applicable to the calendar information (e.g.: Paris, GMT+1).
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Characteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.

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valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CharacteristicRelationship sub-resource fields

id	A String. Unique identifier of the characteristic.
relationshipType	A String. The type of relationship.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ExternalIdentifier sub-resource fields

externalIdentifierType	A String. Type of the identification, typically would be the type of the entity within the external system.
id	A String. Identification of the entity within the external system.
owner	A String. Name of the external system that owns the entity.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Feature sub-resource fields

featureCharacteristic	A Characteristic. This is a list of Characteristics for a particular feature.
featureRelationship	A FeatureRelationship.
id	A String. Unique identifier.
isBundle	A Boolean. True if this is a feature group. Default is false.
isEnabled	A Boolean. True if this feature is enabled. Default is true.
name	A String. This is the name for the feature.
policyConstraint	A PolicyRef. Reference to mangage a list of feature specification policy constraints.
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FeatureRelationship sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
relationshipType	A String. This is the type of the feature relationship.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FloatCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Float. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicAddress sub-resource fields

city	A String. City that the address is in.
country	A String. Country that the address is in.

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countryCode	A StandardIdentifier. Country codes are short alphabetic or numeric geographical codes (geocodes) developed to represent countries and dependent areas, for use in data processing and communications. Several different systems have been developed to do this. The term country code frequently refers to ISO 3166-1 alpha-2 or international dialing codes, the E.164 country calling codes.
externalIdentifier	An ExternalIdentifier.
geographicAddressType	A String. Classification of the address, e.g., residential, industrial.
geographicLocation	A GeographicLocationRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the GeographicLocation entity and not the GeographicLocationRefOrValue class itself.
geographicSubAddress	A GeographicSubAddress.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
locality	A String. An area of defined or undefined boundaries within a local authority or other legislatively defined area, usually rural or semi rural in nature. [ANZLIC-STREET], or a suburb, a bounded locality within a city, town or shire principally of urban character [ANZLICSTREET].
postcode	A String. Descriptor for a postal delivery area, used to speed and simplify the delivery of mail (also know as zipcode).
stateOrProvince	A String. The State or Province that the address is in.
streetName	A String. Name of the street or other street type.
streetNr	A String. Number identifying a specific property on a public street. It may be combined with streetNrLast for ranged addresses.
streetNrLast	A String. Last number in a range of street numbers allocated to a property.
streetNrLastSuffix	A String. Last street number suffix for a ranged address.
streetNrSuffix	A String. The first street number suffix.
streetSuffix	A String. A modifier denoting a relative direction.
streetType	A String. Alley, avenue, boulevard, brae, crescent, drive, highway, lane, terrace, parade, place, tarn, way, wharf.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicLocation sub-resource fields

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bbox	A Number. A bounding box array that contains the geometry. The axes order follows the axes order of the geometry.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicLocationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicSite sub-resource fields

calendar	A CalendarPeriod.
code	A String. A code that may be used for some addressing schemes eg: [ANSI T1.253-1999].
creationDate	A DateTime. Date and time when the GeographicSite was created.
description	A String. Text describing additional information regarding the site.
externalIdentifier	An ExternalIdentifier.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
place	A PlaceRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Place entity and not the PlaceRefOrValue class itself.
relatedParty	A RelatedPartyRefOrPartyRoleRef.
siteRelationship	A GeographicSiteRelationship.
status	A String. The condition of the GeographicSite, such as planned, underConstruction, cancelled, active, inactive, former.

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@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicSiteRelationship sub-resource fields

href	A String. Reference of the related geographic site.
id	A String. Unique identifier of the related site entity within the server.
relationshipType	A String. Type of relationship.
role	A String. Role of the related site in the relationship.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicSubAddress sub-resource fields

buildingName	A String. Allows for buildings that have well-known names.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
levelNumber	A String. Used where a level type may be repeated e.g. BASEMENT 1, BASEMENT 2.
levelType	A String. Describes level types within a building.
name	A String. Name of the subAddress to identify it with a meaningful identification.
privateStreetName	A String. Private streets internal to a property (e.g. a university) may have internal names that are not recorded by the land title office.
privateStreetNumber	A String. Private streets numbers internal to a private street.
subAddressType	A String. Type of subAddress : it can be a subunit or a private street.
subUnit	A GeographicSubAddressUnit. Representation of a SubUnit. It is used for describing subunit within a subAddress e.g. BERTH, FLAT, PIER, SUITE, SHOP, TOWER, UNIT, WHARF.
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

GeographicSubAddressUnit sub-resource fields

subUnitNumber	A String. The discriminator used for the subunit, often just a simple number but may also be a range.
subUnitType	A String. The type of subunit e.g.BERTH, FLAT, PIER, SUITE, SHOP, TOWER, UNIT, WHARF, RACK.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

HourPeriod sub-resource fields

endHour	A String. The time when the status ends applying.
startHour	A String. The time when the status starts applying.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An Integer.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntegerCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship.

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id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An Integer. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntentRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

LogicalResource sub-resource fields

activationFeature	A Feature. Configuration features.
administrativeState	A ResourceAdministrativeStateType. ResourceAdministrativeStateType enumerations.
attachment	An AttachmentRef.
category	A String. Category of the concrete resource. e.g Gold, Silver for MSISDN concrete resource.
description	A String. Free-text description of the resource.
endOperatingDate	A DateTime. A date time(DateTime). The date till the resource is operating.
externalIdentifier	An ExternalIdentifier. An identification of this resource that is owned by or originates in a software system different from the current system. The structure identifies the system itself, the nature of the resource within the system and the unique ID of the resource within the system. It is anticipated that multiple external IDs can be held for a single resource, e.g. if the resource passed through multiple systems on the way to the current system.
href	A String. Hyperlink reference.

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id	A String. Unique identifier.
intent	An IntentRef. Intent reference, for when Intent is used by other entities.
name	A String. The name of the resource.
note	A Note.
operationalState	A ResourceOperationalStateType. ResourceOperationalStateType enumerations.
place	A RelatedPlaceRefOrValue.
relatedParty	A RelatedPartyRefOrPartyRoleRef.
resourceCharacteristic	A Characteristic.
resourceOrderItem	A RelatedResourceOrderItem. A list of resource order items related to this resource.
resourceRelationship	A ResourceRelationship.
resourceSpecification	A ResourceSpecificationRef.
resourceStatus	A ResourceStatusType. ResourceStatusType enumerations.
resourceVersion	A String. A field that identifies the specific version of an instance of a resource.
startOperatingDate	A DateTime. A date time(DateTime). The date from which the resource is operating.
usageState	A ResourceUsageStateType. ResourceUsageStateType enumerations.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
value	A String. The value of the logical resource. E.g '0746712345' for MSISDN's.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Note sub-resource fields

author	A String. Author of the note.
date	A DateTime. Date of the note.
id	A String. Identifier of the note within its containing entity.
text	A String. Text of the note.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Number.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

NumberCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A Number. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ObjectArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An object.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

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ObjectCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	An object. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRoleRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
partyId	A String. The identifier of the engaged party that is linked to the PartyRole object.
partyName	A String. The name of the engaged party that is linked to the PartyRole object.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String. When sub-classing, this defines the sub-class Extensible name.

Place sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PlaceRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PolicyRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyRefOrPartyRoleRef sub-resource fields

partyOrPartyRole	A PartyRefOrPartyRoleRef.

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role	A String. Role played by the related party or party role in the context of the specific entity it is linked to. Such as 'initiator', 'customer', 'salesAgent', 'user'.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPlaceRefOrValue sub-resource fields

place	A PlaceRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Place entity and not the PlaceRefOrValue class itself.
role	A String.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedResourceOrderItem sub-resource fields

itemAction	An OrderItemActionType. Action to be performed on the product.
itemId	A String. Identifier of the order item where the resource was managed.
resourceOrderHref	A String. Reference of the related entity.
resourceOrderId	A String. Unique identifier of a related entity.
role	A String. Role of the resource order item for this resource.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Resource sub-resource fields

activationFeature	A Feature. Configuration features.
administrativeState	A ResourceAdministrativeStateType. ResourceAdministrativeStateType enumerations.
attachment	An AttachmentRef.

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category	A String. Category of the concrete resource. e.g Gold, Silver for MSISDN concrete resource.
description	A String. Free-text description of the resource.
endOperatingDate	A DateTime. A date time(DateTime). The date till the resource is operating.
externalIdentifier	An ExternalIdentifier. An identification of this resource that is owned by or originates in a software system different from the current system. The structure identifies the system itself, the nature of the resource within the system and the unique ID of the resource within the system. It is anticipated that multiple external IDs can be held for a single resource, e.g. if the resource passed through multiple systems on the way to the current system.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intent	An IntentRef. Intent reference, for when Intent is used by other entities.
name	A String. The name of the resource.
note	A Note.
operationalState	A ResourceOperationalStateType. ResourceOperationalStateType enumerations.
place	A RelatedPlaceRefOrValue.
relatedParty	A RelatedPartyRefOrPartyRoleRef.
resourceCharacteristic	A Characteristic.
resourceOrderItem	A RelatedResourceOrderItem. A list of resource order items related to this resource.
resourceRelationship	A ResourceRelationship.
resourceSpecification	A ResourceSpecificationRef.
resourceStatus	A ResourceStatusType. ResourceStatusType enumerations.
resourceVersion	A String. A field that identifies the specific version of an instance of a resource.
startOperatingDate	A DateTime. A date time(DateTime). The date from which the resource is operating.
usageState	A ResourceUsageStateType. ResourceUsageStateType enumerations.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

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ResourceRelationship sub-resource fields

relationshipType	A String. Type of the resource relationship, such as [bundled] if the resource is a bundle and you want to describe the bundled resources inside this bundle; [reliesOn] if the resource needs another already owned resource to rely on (eg: an option on an already owned mobile access resource) [targets] or [isTargeted] (depending on the way of expressing the link) for any other kind of links that may be useful.
resource	A ResourceRefOrValue. The polymorphic attributes @type, @schemaLocation & @referredType are related to the Resource entity and not the ResourceRefOrValue class itself.
resourceRelationshipCharacteristic	A Characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String. Resource Specification version.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StandardIdentifier sub-resource fields

format	A String. Standard/Regulatory definition identifier. e.g., ISO 3166-1.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
value	A String. The value of the resource in the corresponding standard.e.g., a country code value.
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringArrayCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A String.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

StringCharacteristic sub-resource fields

characteristicRelationship	A CharacteristicRelationship.
id	A String. Unique identifier of the characteristic.
name	A String. Name of the characteristic.
value	A String. Value of the characteristic.
valueType	A String. Data type of the value of the characteristic.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Json representation sample(s)

We provide below a JSON representation as example of the 'CloudApplication' resource object.

```
{
    "id": "2287",
    "href": "http://server:port/privateOptimizedBinding/v5/cloudApplication/2287",
    "operationalState": "enable",
    "name": "Security Application ABC",
    "note": [{
            "text": "this is a security application supporting TMF759 binding",
            "@type": "Note"
        }],
```

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```
    "resourceCharacteristic": [{
            "name": "cloudProvider",
            "value": "MS",
            "@type" : "StringCharacteristic"
        },
        {
            "name": "cloudRegion",
            "value": "eastus",
            "@type" : "StringCharacteristic"
        },
        {
            "name": "cloudReference",
            "value": "/subscriptions/00000000-0000-0000-0000-
000000000000/resourceGroups/vnet-
rg/providers/Microsoft.Compute/virtualMachines/vmX",
            "@type" : "StringCharacteristic"
        }],
    "resourceSpecification" : {
        "id" : "11140",
        "href" :
"http://server:port/privateOptimizedBinding/v5/cloudApplicationSpecification/11140",
        "@type" : "ResourceSpecificationRef"
    },
    "@type": "CloudApplication",
    "@schemaLocation": "http://server:port/CloudApplication.schema.json",
    "@baseType": "LogicalResource"
}
```


UserEquipmentSpecification resource

Resource model

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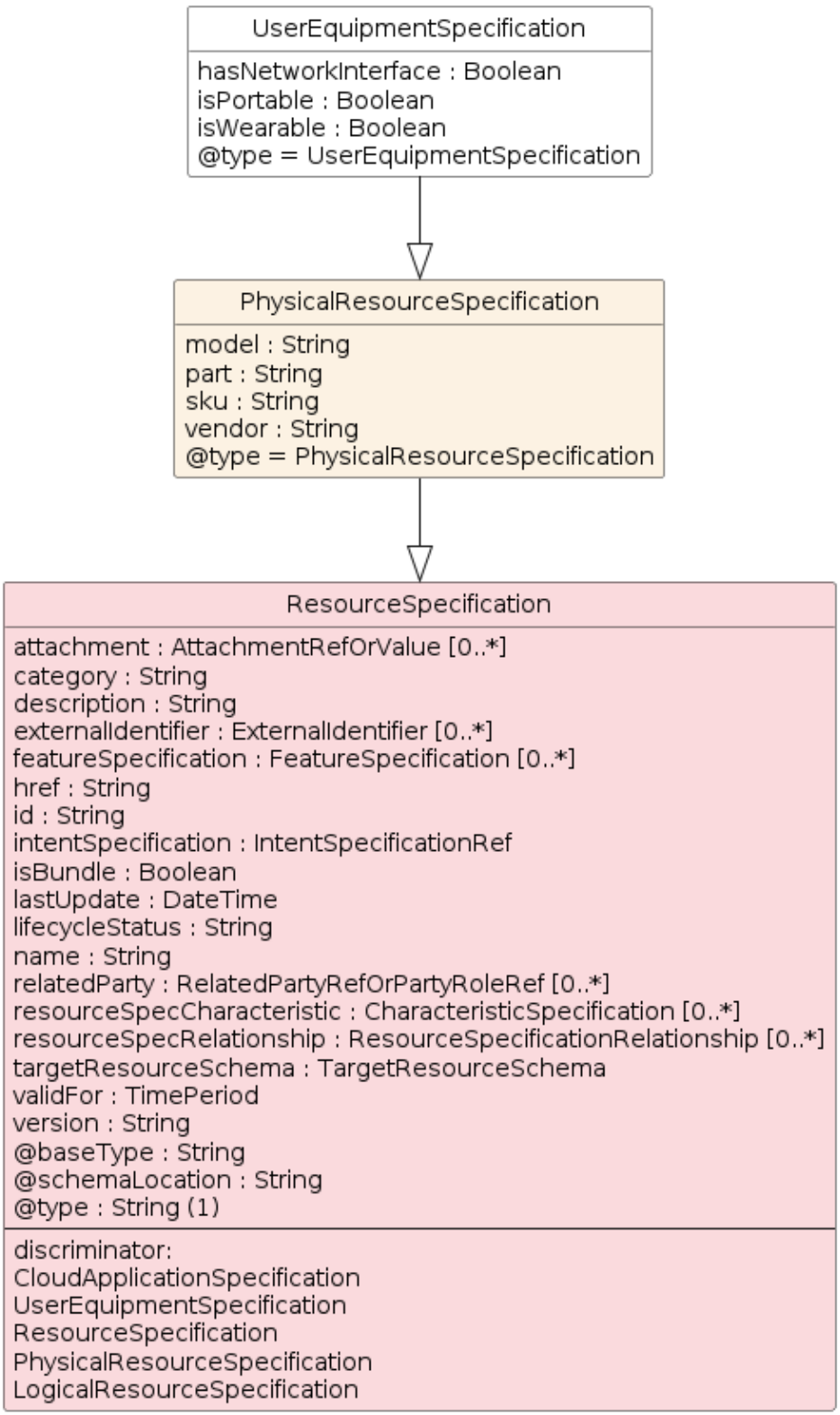


Figure 11 - UserEquipmentSpecification

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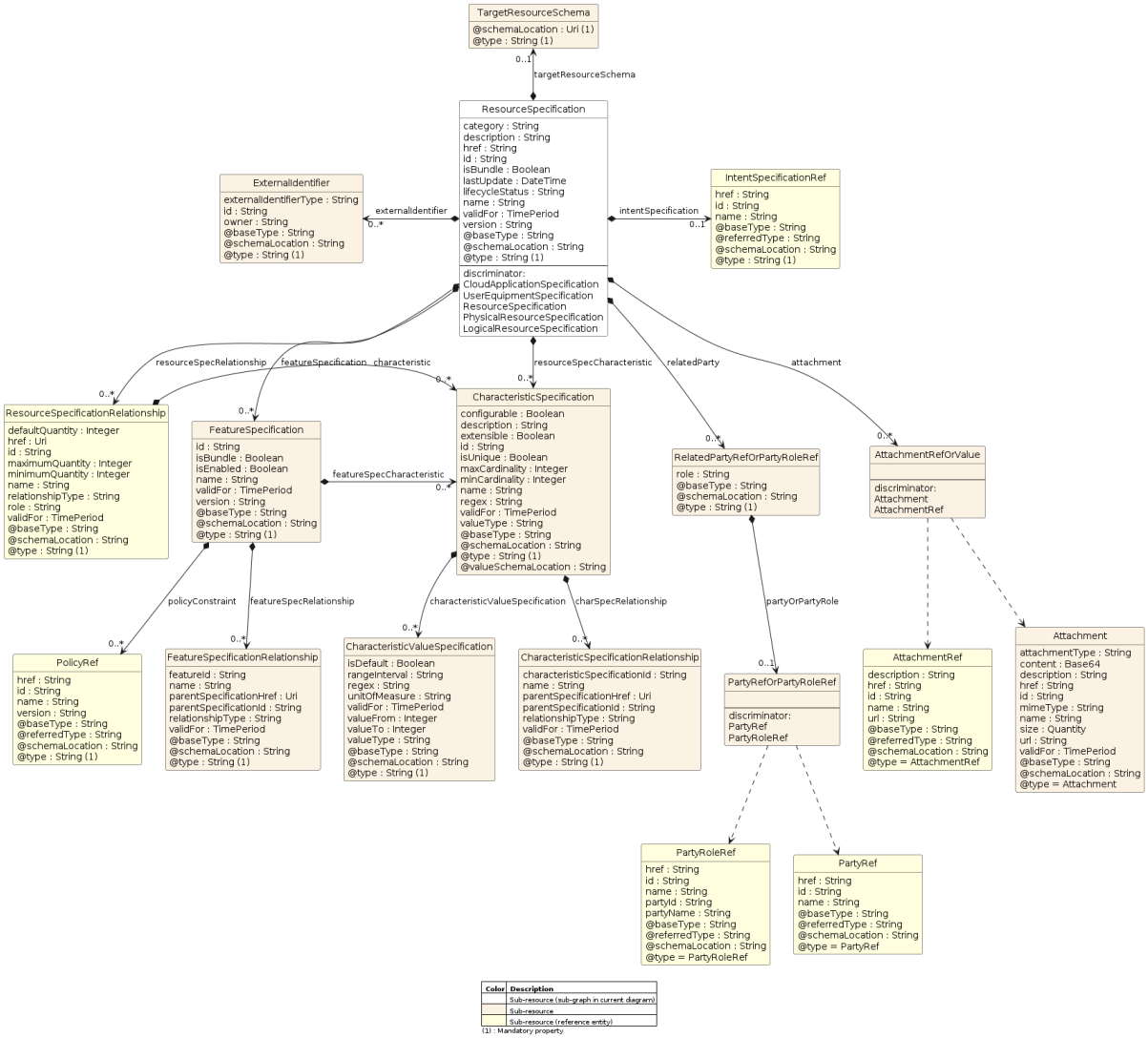


Figure 12 - ResourceSpecification

Field descriptions

UserEquipmentSpecification fields

attachment	An AttachmentRefOrValue. Complements the description of an element (for instance a resource) through video, pictures …
category	A String. Category of the target resource like NetworkConnectivity, PhysicalLinks, Generic, L2Network and so on.
description	A String. Description of this REST resource.
externalIdentifier	An ExternalIdentifier.
featureSpecification	A FeatureSpecification. A list of Features for this specification.
hasNetworkInterface	A Boolean. Tell if UserEquipment has network interface(s).
href	A String. Hyperlink reference.
id	A String. Unique identifier.

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intentSpecification	An IntentSpecificationRef.
isBundle	A Boolean. A flag indicates that if this resource specification is a bundled specification (true) or single (false).
isPortable	A Boolean. Tell if UserEquipment is a portable/movable device.
isWearable	A Boolean. Tell if UserEquipment is a wearable device.
lastUpdate	A DateTime. Date and time of the last update of this REST resource.
lifecycleStatus	A String. Used to indicate the current lifecycle status of the resource specification.
model	A String. This is a string that represents a manufacturer-allocated number used to identify the general type and/or category of the hardware item. This, in combination with the Part and the Vendor, identify different types of hardware items. The SerialNumber can then be used to differentiate between different instances of the same type of hardware item. This is an optional attribute.
name	A String. Name given to this REST resource.
part	A String. This is a string that defines a manufacturer-allocated part number assigned by the organization that manufactures the hardware item. This, in combination with the Model and the Vendor, identify different types of hardware items. The SerialNumber can then be used to differentiate between different instances of the same type of hardware item. This is a REQUIRED attribute.
relatedParty	A RelatedPartyRefOrPartyRoleRef. A related party defines party or party role linked to a specific entity.
resourceSpecCharacteristic	A CharacteristicSpecification. A characteristic quality or distinctive feature of a ResourceSpecification. The characteristic can be take on a discrete value, such as color, can take on a range of values, (for example, sensitivity of 100-240 mV), or can be derived from a formula (for example, usage time (hrs) = 30 - talk time *3). Certain characteristics, such as color, may be configured during the ordering or some other process.
resourceSpecRelationship	A ResourceSpecificationRelationship. A migration, substitution, dependency or exclusivity relationship between/among resource specifications.
sku	A String. This is a string that defines the manufacturer-allocated Stock Keeping Unit (SKU) number of the hardware item. This is an optional attribute.
targetResourceSchema	A TargetResourceSchema. The reference object to the schema and type of target resource which is described by resource specification.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.

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vendor	A String. This is a string that defines the name of the manufacturer. This, in combination with the Model and the Part, identify different types of hardware items. The SerialNumber can then be used to differentiate between different instances of the same type of hardware item. This is a REQUIRED attribute for a physical resource.
version	A String. Resource Specification version.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

Attachment sub-resource fields

attachmentType	A String. A business characterization of the purpose of the attachment, for example logo, instructionManual, contractCopy.
content	A Base64. The actual contents of the attachment object, if embedded, encoded as base64.
description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
mimeType	A String. A technical characterization of the attachment content format using IETF Mime Types.
name	A String. The name of the attachment.
size	A Quantity. An amount in a given unit.
url	A String. Uniform Resource Locator, is a web page address (a subset of URI).
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

AttachmentRef sub-resource fields

description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.

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name	A String. Name of the referred entity.
url	A String. Link to the attachment media/content.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CharacteristicSpecification sub-resource fields

charSpecRelationship	A CharacteristicSpecificationRelationship. An aggregation, migration, substitution, dependency or exclusivity relationship between/among Specification Characteristics.
characteristicValueSpecification	A CharacteristicValueSpecification. A CharacteristicValueSpecification object is used to define a set of attributes, each of which can be assigned to a corresponding set of attributes in a CharacteristicSpecification object. The values of the attributes in the CharacteristicValueSpecification object describe the values of the attributes that a corresponding Characteristic object can take on.
configurable	A Boolean. If true, the Boolean indicates that the target Characteristic is configurable.
description	A String. A narrative that explains the CharacteristicSpecification.
extensible	A Boolean. An indicator that specifies that the values for the characteristic can be extended by adding new values when instantiating a characteristic for a resource.
id	A String. Unique ID for the characteristic.
isUnique	A Boolean. Specifies if the value of this characteristic is unique across all entities instantiated from the specification that uses this characteristc. For example, consider a ProductSpecification for a set-top box, with a CharacteristicSpecification cardID. Each set-top box must have a different value for cardID, so this isUnique attribute would be set to true for the characteristic.
maxCardinality	An Integer. The maximum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where five is the value for the maxCardinality.
minCardinality	An Integer. The minimum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where zero is the value for the minCardinality.
name	A String. A word, term, or phrase by which this characteristic specification is known and distinguished from other characteristic specifications.

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regex	A String. A rule or principle represented in regular expression used to derive the value of a characteristic value.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
valueType	A String. A kind of value that the characteristic can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
@valueSchemaLocation	A String. This (optional) field provides a link to the schema describing the value type.

CharacteristicSpecificationRelationship sub-resource fields

characteristicSpecificationId	A String. Unique identifier of the characteristic within the specification.
name	A String. Name of the target characteristic within the specification.
parentSpecificationHref	A Uri. Hyperlink reference to the parent specification containing the target characteristic.
parentSpecificationId	A String. Unique identifier of the parent specification containing the target characteristic.
relationshipType	A String. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.

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unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ExternalIdentifier sub-resource fields

externalIdentifierType	A String. Type of the identification, typically would be the type of the entity within the external system.
id	A String. Identification of the entity within the external system.
owner	A String. Name of the external system that owns the entity.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FeatureSpecification sub-resource fields

featureSpecCharacteristic	A CharacteristicSpecification. This is a list of characteristics for a particular feature.
featureSpecRelationship	A FeatureSpecificationRelationship. A dependency, exclusivity or aggratation relationship between/among feature specifications.
id	A String. Feature identifier.
isBundle	A Boolean. A flag indicating if this is a feature group (true) or not (false).
isEnabled	A Boolean. A flag indicating if the feature is enabled (true) or not (false).
name	A String. This is the name for the feature.

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policyConstraint	A PolicyRef. Reference to mangage a list of feature specification policy constraints.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
version	A String. Version of the feature specification.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FeatureSpecificationRelationship sub-resource fields

featureId	A String. Unique identifier of the target feature specification.
name	A String. This is the name of the target feature specification.
parentSpecificationHref	A Uri. Hyperlink reference to the parent specification containing the target feature.
parentSpecificationId	A String. Unique identifier of the parent specification containing the target feature.
relationshipType	A String. This is the type of the feature specification relationship.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntentSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRef sub-resource fields

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href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRoleRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
partyId	A String. The identifier of the engaged party that is linked to the PartyRole object.
partyName	A String. The name of the engaged party that is linked to the PartyRole object.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PhysicalResourceSpecification sub-resource fields

attachment	An AttachmentRefOrValue. Complements the description of an element (for instance a resource) through video, pictures …
category	A String. Category of the target resource like NetworkConnectivity, PhysicalLinks, Generic, L2Network and so on.
description	A String. Description of this REST resource.
externalIdentifier	An ExternalIdentifier.
featureSpecification	A FeatureSpecification. A list of Features for this specification.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
intentSpecification	An IntentSpecificationRef.

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isBundle	A Boolean. A flag indicates that if this resource specification is a bundled specification (true) or single (false).
lastUpdate	A DateTime. Date and time of the last update of this REST resource.
lifecycleStatus	A String. Used to indicate the current lifecycle status of the resource specification.
model	A String. This is a string that represents a manufacturer-allocated number used to identify the general type and/or category of the hardware item. This, in combination with the Part and the Vendor, identify different types of hardware items. The SerialNumber can then be used to differentiate between different instances of the same type of hardware item. This is an optional attribute.
name	A String. Name given to this REST resource.
part	A String. This is a string that defines a manufacturer-allocated part number assigned by the organization that manufactures the hardware item. This, in combination with the Model and the Vendor, identify different types of hardware items. The SerialNumber can then be used to differentiate between different instances of the same type of hardware item. This is a REQUIRED attribute.
relatedParty	A RelatedPartyRefOrPartyRoleRef. A related party defines party or party role linked to a specific entity.
resourceSpecCharacteristic	A CharacteristicSpecification. A characteristic quality or distinctive feature of a ResourceSpecification. The characteristic can be take on a discrete value, such as color, can take on a range of values, (for example, sensitivity of 100-240 mV), or can be derived from a formula (for example, usage time (hrs) = 30 - talk time *3). Certain characteristics, such as color, may be configured during the ordering or some other process.
resourceSpecRelationship	A ResourceSpecificationRelationship. A migration, substitution, dependency or exclusivity relationship between/among resource specifications.
sku	A String. This is a string that defines the manufacturer-allocated Stock Keeping Unit (SKU) number of the hardware item. This is an optional attribute.
targetResourceSchema	A TargetResourceSchema. The reference object to the schema and type of target resource which is described by resource specification.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
vendor	A String. This is a string that defines the name of the manufacturer. This, in combination with the Model and the Part, identify different types of hardware items. The SerialNumber can then be used to differentiate between different instances of the same type of hardware item. This is a REQUIRED attribute for a physical resource.
version	A String. Resource Specification version.

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@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PolicyRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyRefOrPartyRoleRef sub-resource fields

partyOrPartyRole	A PartyRefOrPartyRoleRef.
role	A String. Role played by the related party or party role in the context of the specific entity it is linked to. Such as 'initiator', 'customer', 'salesAgent', 'user'.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceSpecification sub-resource fields

attachment	An AttachmentRefOrValue. Complements the description of an element (for instance a resource) through video, pictures …
category	A String. Category of the target resource like NetworkConnectivity, PhysicalLinks, Generic, L2Network and so on.
description	A String. Description of this REST resource.
externalIdentifier	An ExternalIdentifier.
featureSpecification	A FeatureSpecification. A list of Features for this specification.
href	A String. Hyperlink reference.

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id	A String. Unique identifier.
intentSpecification	An IntentSpecificationRef.
isBundle	A Boolean. A flag indicates that if this resource specification is a bundled specification (true) or single (false).
lastUpdate	A DateTime. Date and time of the last update of this REST resource.
lifecycleStatus	A String. Used to indicate the current lifecycle status of the resource specification.
name	A String. Name given to this REST resource.
relatedParty	A RelatedPartyRefOrPartyRoleRef. A related party defines party or party role linked to a specific entity.
resourceSpecCharacteristic	A CharacteristicSpecification. A characteristic quality or distinctive feature of a ResourceSpecification. The characteristic can be take on a discrete value, such as color, can take on a range of values, (for example, sensitivity of 100-240 mV), or can be derived from a formula (for example, usage time (hrs) = 30 - talk time *3). Certain characteristics, such as color, may be configured during the ordering or some other process.
resourceSpecRelationship	A ResourceSpecificationRelationship. A migration, substitution, dependency or exclusivity relationship between/among resource specifications.
targetResourceSchema	A TargetResourceSchema. The reference object to the schema and type of target resource which is described by resource specification.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
version	A String. Resource Specification version.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceSpecificationRelationship sub-resource fields

characteristic	A CharacteristicSpecification. A characteristic that refines the relationship. For example, consider the relationship between a slot and a card. For a half-height card it is important to know the position at which the card is inserted, so a characteristic Position might be defined on the relationship to allow capturing of this in the inventory.
defaultQuantity	An Integer. The default number of the related resource that should be instantiated, for example a rack would typically have 4 cards, although it could support more.
href	A Uri. Reference of the target ResourceSpecification.

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id	A String. Unique identifier of target ResourceSpecification.
maximumQuantity	An Integer. The maximum number of the related resource that should be instantiated, for example a rack supports a maximum of 16 cards.
minimumQuantity	An Integer. The minimum number of the related resource that should be instantiated, for example a rack must have at least 1 card.
name	A String. The name given to the target resource specification instance.
relationshipType	A String. Type of relationship such as migration, substitution, dependency, exclusivity.
role	A String. The association role for this resource specification.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

TargetResourceSchema sub-resource fields

@schemaLocation	A Uri. This field provides a link to the schema describing the target resource.
@type	A String. Class type of the target resource.

Json representation sample(s)

We provide below a JSON representation as example of the 'UserEquipmentSpecification' resource object.

```
{
    "id": "11140",
    "href":
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11140",
    "description": "This is an example of a user equipment specification that we
want for or binded UE.",
    "version": "1.0",
    "isPortable": false,
    "isWearable": false,
    "hasNetworkInterface": true,
    "vendor": "MyUEVendor",
    "model": "MyUEModel",
    "name": "MySpecificationName",
    "isBundle": false,
    "resourceSpecCharacteristic":[
        {
            "name": "MSISDN",
            "valueType": "text",
            "@type": "CharacteristicSpecification"
```

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```
        }
    ],
    "@type": "UserEquipmentSpecification",
    "@schemaLocation": "http://server:port/UserEquipmentSpecification.schema.json",
    "@baseType": "PhysicalResourceSpecification"
}
```

We provide below a JSON representation as example of the 'UserEquipmentSpecification' resource object.

```
{
    "id": "11143",
    "href":
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11143",
    "description": "This is an example of a user equipment specification that we
want for or binded UE.",
    "version": "1.0",
    "isPortable": false,
    "isWearable": false,
    "hasNetworkInterface": true,
    "vendor": "MyUEVendor",
    "model": "MyUEModel",
    "name": "MySpecificationName",
    "isBundle": false,
    "featureSpecification":[
        {
            "name": "privateOptimizedBindingFeature",
            "featureSpecRelationship" :[
                {
                    "relationshipType" : "requires",
                    "name" : "privateOptimizedBindingFeature",
                    "@type" : "FeatureSpecificationRelationship"
                }
            ],
            "featureSpecCharacteristic" : [
                {
                    "name": "privateOptimizedBindingCloudAppHref",
                    "valueType": "text",
                    "@type": "CharacteristicSpecification"
                }
            ],
            "@type": "FeatureSpecification"
        }
    ],
    "resourceSpecCharacteristic":[
        {
            "name": "MSISDN",
            "valueType": "text",
            "@type": "CharacteristicSpecification"}
    ],
    "@type": "UserEquipmentSpecification",
    "@schemaLocation": "http://server:port/UserEquipmentSpecification.schema.json",
    "@baseType": "PhysicalResourceSpecification"
}
```

CloudApplicationSpecification resource

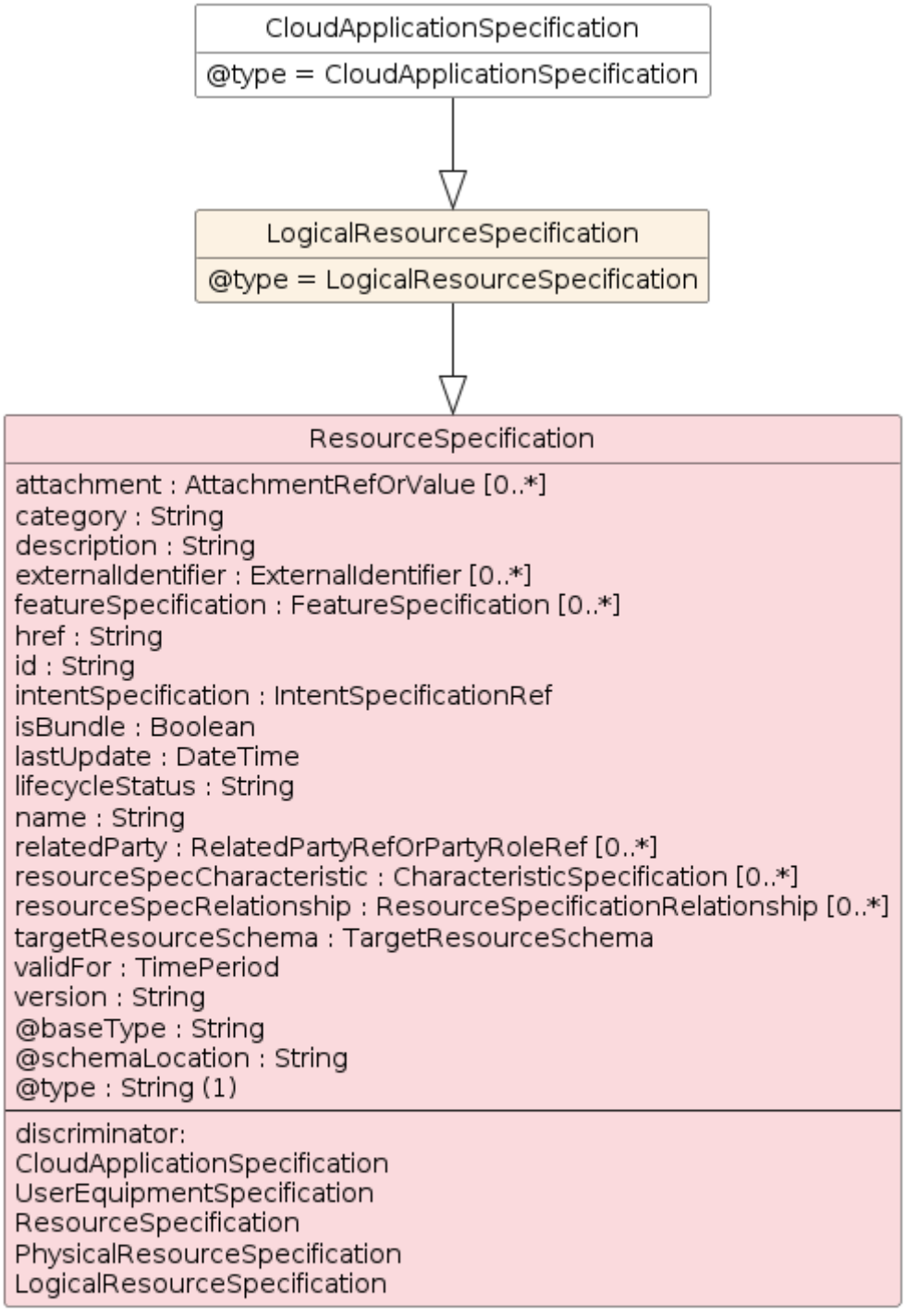
This is a derived class of LogicalResourceSpecification, and is used to define the invariant

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characteristics and behavior (attributes, methods, constraints, and relationships) of a CloudApplication.

Resource model

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Color	Description
	Resource (entry point)
	Sub-resource
	Sub-resource with details in separate diagram

(1) : Mandatory property

Figure 13 - CloudApplicationSpecification

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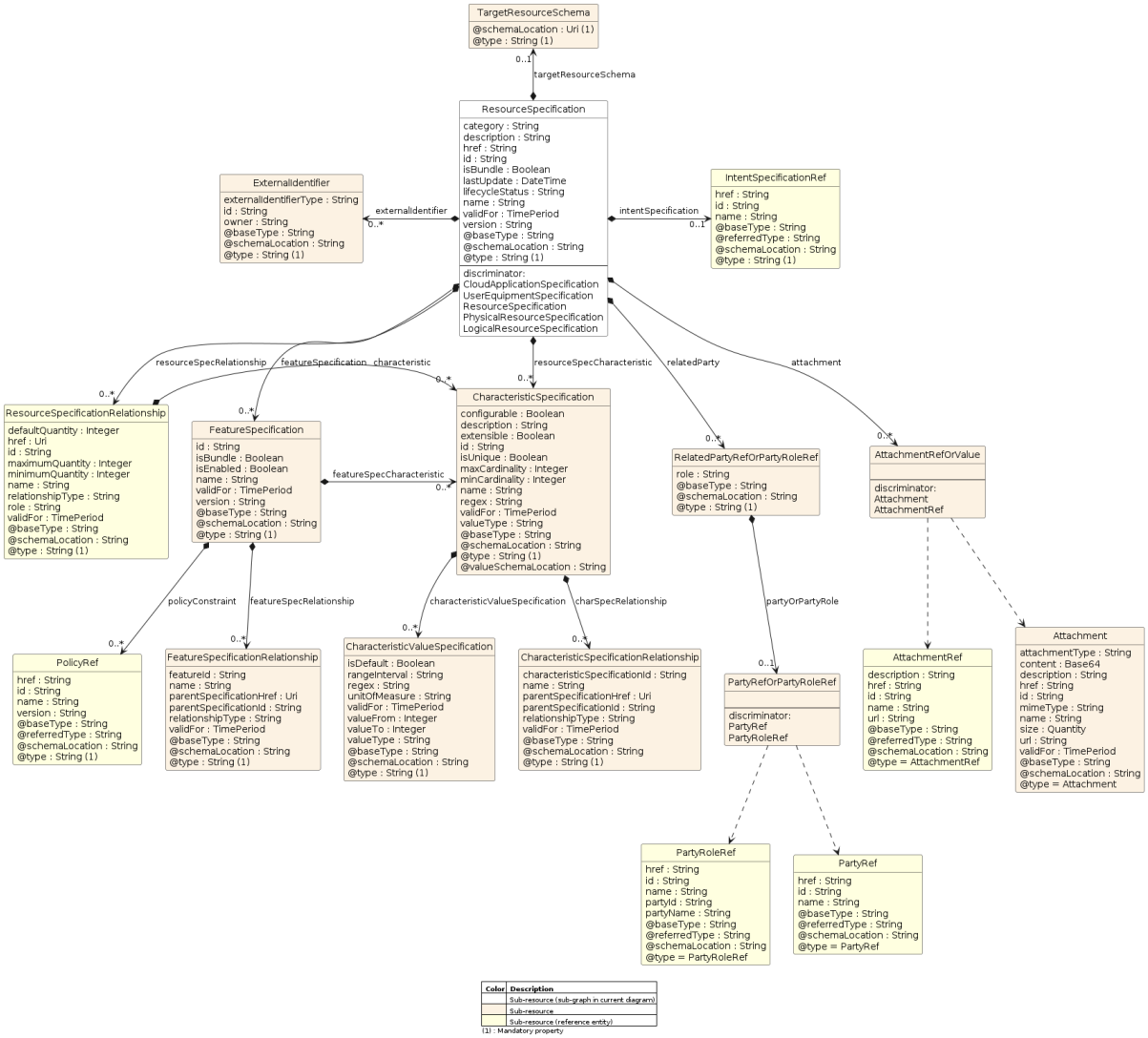


Figure 14 - ResourceSpecification

Field descriptions

Attachment sub-resource fields

attachmentType	A String. A business characterization of the purpose of the attachment, for example logo, instructionManual, contractCopy.
content	A Base64. The actual contents of the attachment object, if embedded, encoded as base64.
description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
mimeType	A String. A technical characterization of the attachment content format using IETF Mime Types.
name	A String. The name of the attachment.

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size	A Quantity. An amount in a given unit.
url	A String. Uniform Resource Locator, is a web page address (a subset of URI).
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

AttachmentRef sub-resource fields

description	A String. A narrative text describing the content of the attachment.
href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
url	A String. Link to the attachment media/content.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CharacteristicSpecification sub-resource fields

charSpecRelationship	A CharacteristicSpecificationRelationship. An aggregation, migration, substitution, dependency or exclusivity relationship between/among Specification Characteristics.
characteristicValueSpecification	A CharacteristicValueSpecification. A CharacteristicValueSpecification object is used to define a set of attributes, each of which can be assigned to a corresponding set of attributes in a CharacteristicSpecification object. The values of the attributes in the CharacteristicValueSpecification object describe the values of the attributes that a corresponding Characteristic object can take on.
configurable	A Boolean. If true, the Boolean indicates that the target Characteristic is configurable.
description	A String. A narrative that explains the CharacteristicSpecification.

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extensible	A Boolean. An indicator that specifies that the values for the characteristic can be extended by adding new values when instantiating a characteristic for a resource.
id	A String. Unique ID for the characteristic.
isUnique	A Boolean. Specifies if the value of this characteristic is unique across all entities instantiated from the specification that uses this characteristc. For example, consider a ProductSpecification for a set-top box, with a CharacteristicSpecification cardID. Each set-top box must have a different value for cardID, so this isUnique attribute would be set to true for the characteristic.
maxCardinality	An Integer. The maximum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where five is the value for the maxCardinality.
minCardinality	An Integer. The minimum number of instances a CharacteristicValue can take on. For example, zero to five phone numbers in a group calling plan, where zero is the value for the minCardinality.
name	A String. A word, term, or phrase by which this characteristic specification is known and distinguished from other characteristic specifications.
regex	A String. A rule or principle represented in regular expression used to derive the value of a characteristic value.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
valueType	A String. A kind of value that the characteristic can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.
@valueSchemaLocation	A String. This (optional) field provides a link to the schema describing the value type.

CharacteristicSpecificationRelationship sub-resource fields

characteristicSpecificationId	A String. Unique identifier of the characteristic within the specification.
name	A String. Name of the target characteristic within the specification.
parentSpecificationHref	A Uri. Hyperlink reference to the parent specification containing the target characteristic.
parentSpecificationId	A String. Unique identifier of the parent specification containing the target characteristic.

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relationshipType	A String. Type of relationship such as aggregation, migration, substitution, dependency, exclusivity.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

CharacteristicValueSpecification sub-resource fields

isDefault	A Boolean. If true, the Boolean Indicates if the value is the default value for a characteristic.
rangeInterval	A String. An indicator that specifies the inclusion or exclusion of the valueFrom and valueTo attributes. If applicable, possible values are "open", "closed", "closedBottom" and "closedTop".
regex	A String. A regular expression constraint for given value.
unitOfMeasure	A String. A length, surface, volume, dry measure, liquid measure, money, weight, time, and the like. In general, a determinate quantity or magnitude of the kind designated, taken as a standard of comparison for others of the same kind, in assigning to them numerical values, as 1 foot, 1 yard, 1 mile, 1 square foot.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
valueFrom	An Integer. The low range value that a characteristic can take on.
valueTo	An Integer. The upper range value that a characteristic can take on.
valueType	A String. A kind of value that the characteristic value can take on, such as numeric, text and so forth.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ExternalIdentifier sub-resource fields

externalIdentifierType	A String. Type of the identification, typically would be the type of the entity within the external system.
id	A String. Identification of the entity within the external system.
owner	A String. Name of the external system that owns the entity.
@baseType	A String. When sub-classing, this defines the super-class.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FeatureSpecification sub-resource fields

featureSpecCharacteristic	A CharacteristicSpecification. This is a list of characteristics for a particular feature.
featureSpecRelationship	A FeatureSpecificationRelationship. A dependency, exclusivity or aggratation relationship between/among feature specifications.
id	A String. Feature identifier.
isBundle	A Boolean. A flag indicating if this is a feature group (true) or not (false).
isEnabled	A Boolean. A flag indicating if the feature is enabled (true) or not (false).
name	A String. This is the name for the feature.
policyConstraint	A PolicyRef. Reference to mangage a list of feature specification policy constraints.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
version	A String. Version of the feature specification.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

FeatureSpecificationRelationship sub-resource fields

featureId	A String. Unique identifier of the target feature specification.
name	A String. This is the name of the target feature specification.
parentSpecificationHref	A Uri. Hyperlink reference to the parent specification containing the target feature.
parentSpecificationId	A String. Unique identifier of the parent specification containing the target feature.
relationshipType	A String. This is the type of the feature specification relationship.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.

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@type	A String. When sub-classing, this defines the sub-class Extensible name.

IntentSpecificationRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PartyRoleRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
partyId	A String. The identifier of the engaged party that is linked to the PartyRole object.
partyName	A String. The name of the engaged party that is linked to the PartyRole object.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.

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@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

PolicyRef sub-resource fields

href	A String. Hyperlink reference.
id	A String. Unique identifier.
name	A String. Name of the referred entity.
version	A String.
@baseType	A String. When sub-classing, this defines the super-class.
@referredType	A String. The actual type of the target instance when needed for disambiguation.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

RelatedPartyRefOrPartyRoleRef sub-resource fields

partyOrPartyRole	A PartyRefOrPartyRoleRef.
role	A String. Role played by the related party or party role in the context of the specific entity it is linked to. Such as 'initiator', 'customer', 'salesAgent', 'user'.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceSpecification sub-resource fields

attachment	An AttachmentRefOrValue. Complements the description of an element (for instance a resource) through video, pictures …
category	A String. Category of the target resource like NetworkConnectivity, PhysicalLinks, Generic, L2Network and so on.
description	A String. Description of this REST resource.
externalIdentifier	An ExternalIdentifier.
featureSpecification	A FeatureSpecification. A list of Features for this specification.
href	A String. Hyperlink reference.
id	A String. Unique identifier.

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intentSpecification	An IntentSpecificationRef.
isBundle	A Boolean. A flag indicates that if this resource specification is a bundled specification (true) or single (false).
lastUpdate	A DateTime. Date and time of the last update of this REST resource.
lifecycleStatus	A String. Used to indicate the current lifecycle status of the resource specification.
name	A String. Name given to this REST resource.
relatedParty	A RelatedPartyRefOrPartyRoleRef. A related party defines party or party role linked to a specific entity.
resourceSpecCharacteristic	A CharacteristicSpecification. A characteristic quality or distinctive feature of a ResourceSpecification. The characteristic can be take on a discrete value, such as color, can take on a range of values, (for example, sensitivity of 100-240 mV), or can be derived from a formula (for example, usage time (hrs) = 30 - talk time *3). Certain characteristics, such as color, may be configured during the ordering or some other process.
resourceSpecRelationship	A ResourceSpecificationRelationship. A migration, substitution, dependency or exclusivity relationship between/among resource specifications.
targetResourceSchema	A TargetResourceSchema. The reference object to the schema and type of target resource which is described by resource specification.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
version	A String. Resource Specification version.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

ResourceSpecificationRelationship sub-resource fields

characteristic	A CharacteristicSpecification. A characteristic that refines the relationship. For example, consider the relationship between a slot and a card. For a half-height card it is important to know the position at which the card is inserted, so a characteristic Position might be defined on the relationship to allow capturing of this in the inventory.
defaultQuantity	An Integer. The default number of the related resource that should be instantiated, for example a rack would typically have 4 cards, although it could support more.
href	A Uri. Reference of the target ResourceSpecification.
id	A String. Unique identifier of target ResourceSpecification.

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maximumQuantity	An Integer. The maximum number of the related resource that should be instantiated, for example a rack supports a maximum of 16 cards.
minimumQuantity	An Integer. The minimum number of the related resource that should be instantiated, for example a rack must have at least 1 card.
name	A String. The name given to the target resource specification instance.
relationshipType	A String. Type of relationship such as migration, substitution, dependency, exclusivity.
role	A String. The association role for this resource specification.
validFor	A TimePeriod. A period of time, either as a deadline (endDateTime only) a startDateTime only, or both.
@baseType	A String. When sub-classing, this defines the super-class.
@schemaLocation	A String. A URI to a JSON-Schema file that defines additional attributes and relationships.
@type	A String. When sub-classing, this defines the sub-class Extensible name.

TargetResourceSchema sub-resource fields

@schemaLocation	A Uri. This field provides a link to the schema describing the target resource.
@type	A String. Class type of the target resource.

Json representation sample(s)

We provide below a JSON representation as example of the 'CloudApplicationSpecification' resource object.

```
{
    "id": "11140",
    "href":
"http://server:port/privateOptimizedBinding/v5/cloudApplicationSpecification/11140",
    "description": "This is an example of a cloud application specification.",
    "version": "1.0",
    "name": "MyAppSpecificationName",
    "resourceSpecCharacteristic":[
        {
            "name": "cloudProvider",
            "valueType": "text",
            "@type": "CharacteristicSpecification"
        },
        {
            "name": "cloudRegion",
            "valueType": "text",
            "@type": "CharacteristicSpecification"
        },
        {
            "name": "cloudReference",
            "valueType": "text",
```

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```
            "@type": "CharacteristicSpecification"
        }
    ],
    "@type": "CloudApplicationSpecification",
    "@schemaLocation":
"http://server:port/CloudApplicationSpecification.schema.json",
    "@baseType": "LogicalResourceSpecification"
}
```

Notification Resource Models

14 notifications are defined for this API.

Notifications related to UserEquipment:

- User Equipment Create Event
- User Equipment Attribute Value Change Event
- User Equipment Delete Event
- User Equipment State Change Event

Notifications related to CloudApplication:

- Cloud Application Create Event
- Cloud Application Attribute Value Change Event
- Cloud Application Delete Event
- Cloud Application State Change Event

Notifications related to UserEquipmentSpecification:

- User Equipment Specification Attribute Value Change Event
- User Equipment Specification Create Event
- User Equipment Specification Delete Event

Notifications related to CloudApplicationSpecification:

- Cloud Application Specification Attribute Value Change Event
- Cloud Application Specification Create Event
- Cloud Application Specification Delete Event

The notification structure for all notifications in this API follow the pattern depicted by the figure below. A notification event resource (depicted by "SpecificEvent" placeholder) is a sub class of a generic Event structure containing at least an id of the event occurrence (eventId), an event timestamp (eventTime), and the name of the resource (eventType). This notification structure owns an event payload structure ("SpecificEventPayload" placeholder) linked to the resource concerned by the notification using the resource name as access field ("resourceName" placeholder).

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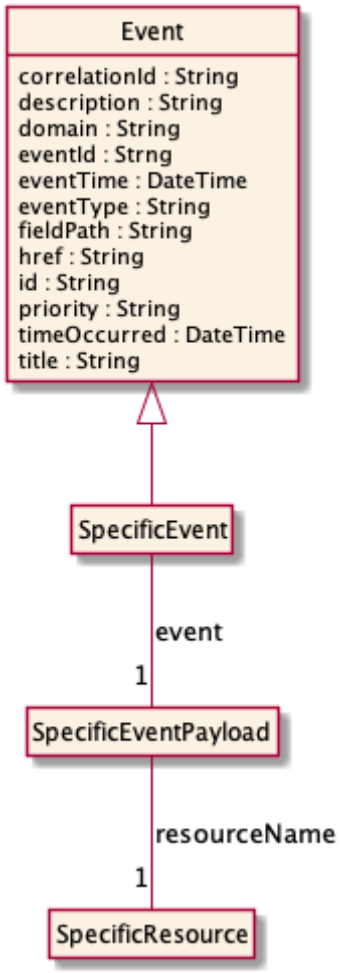


Figure 15 Notification Pattern

User Equipment Create Event

Message example for UserEquipmentCreateEvent event

```
Content-Type: application/json

{
    "correlationId": "126-54f",
    "description": "UserEquipmentCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "125",
    "eventTime": "2021-09-27T07:43:59.059Z",
    "eventType": "UserEquipmentCreateEvent",
    "priority": "1",
    "timeOcurred": "2021-09-27T07:43:59.059Z",
    "title": "UserEquipmentCreateEvent",
    "event": {
      "userEquipment": {
        "id": "42",
        "href": "http://server:port/privateOptimizedBinding/v5/userEquipment/42",
        "description": "This is an example of a user equipment that we want to bind
to a cloud application.",
        "name": "Security Device XYZ",
        "operationalState": "disable",
        "note": [{
```

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```
            "text": "something about this resource",
            "@type": "Note"
        }],
        "place": [{
          "role": "HeadQuarter",
          "place": {
              "id": "2b8cea8f-642d-42e0-bdb2-6f47fabb3f07",
              "href": "https://host/tmf-
api/geographicAddressManagement/v4/geographicAddress/2b8cea8f-642d-42e0-bdb2-
6f47fabb3f07",
              "streetNr": "225",
              "streetNrSuffix": "B",
              "streetNrLast": "",
              "streetNrLastSuffix": "",
              "streetName": " Strathmore",
              "streetType": "Terrace",
              "streetSuffix": "",
              "postcode": "5004",
              "locality": "Brighton.",
              "city": "Brighton",
              "stateOrProvince": "SA",
              "country": "Australia",
              "@type": "GeographicAddress"
          },
          "@type": "RelatedPlaceRefOrValue"
        }],
        "resourceCharacteristic": [{
            "name": "MSISDN",
            "valueType": "string",
            "value": "3345667788643",
            "@type": "StringCharacteristic"
        }],
        "resourceSpecification" : {
            "id": "11140",
            "href":
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11140",
            "@type" : "ResourceSpecificationRef"
        },
        "@type": "UserEquipment",
        "@schemaLocation": "http://server:port/UserEquipment.schema.json",
        "@baseType": "PhysicalResource"
      }
    },
    "relatedParty": [
      {
        "id": "56d",
        "role": "Issuer",
        "@type": "RelatedParty",
        "@referredType": "Organization"
      }
    ],
    "reportingSystem": {
      "id": "123",
      "name": "CRM app",
      "href": "",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "source": {
      "id": "123",
      "name": "CRM app",
      "href": "",
      "@type": "ReportingResource",
```


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```
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "UserEquipmentCreateEvent"
}
```

User Equipment Attribute Value Change Event

Message example for UserEquipmentAttributeValueChangeEvent event

```
Content-Type: application/json

{
    "correlationId": "8a1d7f00-245b",
    "description": "UserEquipmentAttributeValueChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "467c-937d-9afe827e85ee",
    "eventTime": "2022-09-14T12:14:28.774Z",
    "eventType": "UserEquipmentAttributeValueChangeEvent",
    "priority": "3",
    "timeOcurred": "2022-09-14T12:14:23.473Z",
    "title": "UserEquipmentAttributeValueChangeEvent",
    "event": {
        "userEquipment": {
            "href":
"https://host:port/privateOptimizedBinding/v5/UserEquipment/9693",
            "id": "9693",
            "@type": "UserEquipment",
            "name": "User Equipment XYZ",
            "note": [{
                "text": "this is an updated user equipment supporting TMF759
binding",
                "@type": "Note"
            }]
        }
    },
    "reportingSystem": {
        "id": "533",
        "name": "APP-963",
        "href": "",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "200",
        "name": "APP-893",
        "href": "",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "UserEquipmentAttributeValueChangeEvent"
}
```

User Equipment Delete Event

This operation deletes a UserEquipment entity.

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```
Content-Type: application/json

{
    "correlationId": "145d-8hh",
    "description": "UserEquipmentDeleteEvent illustration",
    "domain": "Commercial",
    "eventId": "523",
    "eventTime": "2021-09-27T07:43:59.059Z",
    "eventType": "UserEquipmentDeleteEvent",
    "priority": "3",
    "timeOcurred": "2021-09-27T07:43:59.059Z",
    "title": "UserEquipmentDeleteEvent",
    "event": {
      "userEquipment": {
        "id": "9693",
        "href": "https://host:port/privateOptimizedBinding/v5/userEquipment/9693",
        "@type": "UserEquipment"
      }
    },
    "reportingSystem": {
      "id": "123",
      "name": "Reporting app",
      "href": "",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "source": {
      "id": "123",
      "name": "Reporting app",
      "href": "",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "UserEquipmentDeleteEvent"
}
```

User Equipment State Change Event

Message example for UserEquipmentStateChangeEvent event with ?fields=state property only

```
Content-Type: application/json

{
    "correlationId": "333-fd6",
    "description": "UserEquipmentStateChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "569",
    "eventTime": "2021-09-27T07:43:59.059Z",
    "eventType": "UserEquipmentStateChangeEvent",
    "priority": "1",
    "timeOcurred": "2021-09-27T07:43:59.059Z",
    "title": "UserEquipmentStateChangeEvent",
    "event": {
      "userEquipment": {
        "id": "2287",
        "href": "http://server:port/privateOptimizedBinding/v5/UserEquipment/2287",
```

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```
        "operationalState": "enable",
        "@type": "UserEquipment",
        "@schemaLocation": "http://server:port/UserEquipment.schema.json",
        "@baseType": "PhysicalResource"
      }
    },
    "reportingSystem": {
      "id": "123",
      "name": "CRM app",
      "href": "",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "source": {
      "id": "123",
      "name": "CRM app",
      "href": "",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "UserEquipmentStateChangeEvent"
  }
```


Cloud Application Create Event

Message example for CloudApplicationCreateEvent event


```
Content-Type: application/json

{
    "correlationId": "126-54f",
    "description": "CloudApplicationCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "125",
    "eventTime": "2021-09-27T07:43:59.059Z",
    "eventType": "CloudApplicationCreateEvent",
    "priority": "1",
    "timeOcurred": "2021-09-27T07:43:59.059Z",
    "title": "CloudApplicationCreateEvent",
    "event": {
      "cloudApplication": {
        "id": "2287",
        "href":
"http://server:port/privateOptimizedBinding/v5/cloudApplication/2287",
        "operationalState": "enable",
        "name": "Security Application ABC",
        "note": [{
                "text": "this is a security application supporting TMF759 binding",
                "@type": "Note"
            }],
        "resourceCharacteristic": [{
            "name": "cloudProvider",
            "valueType": "string",
            "value": "MS",
            "@type": "StringCharacteristic"
        },
        {
            "name": "cloudRegion",
            "valueType": "string",
```


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```
            "value": "eastus",
            "@type": "StringCharacteristic"
        },
        {
            "name": "cloudReference",
            "valueType": "string",
            "value": "/subscriptions/00000000-0000-0000-0000-
000000000000/resourceGroups/vnet-
rg/providers/Microsoft.Compute/virtualMachines/vmX",
            "@type": "StringCharacteristic"
        }],
        "resourceSpecification" : {
            "id" : "11140",
            "href" :
"http://server:port/privateOptimizedBinding/v5/cloudApplicationSpecification/11140",
            "@type": "ResourceSpecificationRef"
        },
        "@type": "CloudApplication",
        "@schemaLocation": "http://server:port/CloudApplication.schema.json",
        "@baseType": "LogicalResource"
      }
    },
    "relatedParty": [
      {
        "id": "56d",
        "role": "Issuer",
        "@type": "RelatedParty",
        "@referredType": "Organization"
      }
    ],
    "reportingSystem": {
      "id": "123",
      "name": "CRM app",
      "href": "",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "source": {
      "id": "123",
      "name": "CRM app",
      "href": "",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "CloudApplicationCreateEvent"
}
```


Cloud Application Attribute Value Change Event

Message example for CloudApplicationAttributeValueChangeEvent event


```
Content-Type: application/json

{
    "correlationId": "8a1d7f00-245b",
    "description": "CloudApplicationAttributeValueChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "467c-937d-9afe827e85ee",
    "eventTime": "2022-09-14T12:14:28.774Z",
```


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```
    "eventType": "CloudApplicationAttributeValueChangeEvent",
    "priority": "3",
    "timeOcurred": "2022-09-14T12:14:23.473Z",
    "title": "CloudApplicationAttributeValueChangeEvent",
    "event": {
        "cloudApplication": {
            "href":
"https://host:port/privateOptimizedBinding/v5/CloudApplication/9693",
            "id": "9693",
            "@type": "CloudApplication",
            "name": "Security Application ABC",
            "note": [{
                "text": "this is an updated security application supporting TMF759
binding",
                "@type": "Note"
            }]
        }
    },
    "reportingSystem": {
        "id": "533",
        "name": "APP-963",
        "href": "",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "200",
        "name": "APP-893",
        "href": "",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "CloudApplicationAttributeValueChangeEvent"
}
```


Cloud Application Delete Event

Message example for CloudApplicationDeleteEvent event


```
Content-Type: application/json

{
    "correlationId": "145d-8hh",
    "description": "CloudApplicationDeleteEvent illustration",
    "domain": "Commercial",
    "eventId": "523",
    "eventTime": "2021-09-27T07:43:59.059Z",
    "eventType": "CloudApplicationDeleteEvent",
    "priority": "3",
    "timeOcurred": "2021-09-27T07:43:59.059Z",
    "title": "CloudApplicationDeleteEvent",
    "event": {
      "cloudApplication": {
        "id": "9693",
        "href":
"https://host:port/privateOptimizedBinding/v5/CloudApplication/9693",
        "@type": "CloudApplication"
      }
    },
```


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```
    "reportingSystem": {
      "id": "123",
      "name": "Reporting app",
      "href": "",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "source": {
      "id": "123",
      "name": "Reporting app",
      "href": "",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "CloudApplicationDeleteEvent"
}
```

Cloud Application State Change Event

Message example for CloudApplicationStateChangeEvent event with ?fields=state property only

```
Content-Type: application/json

{
    "correlationId": "333-fd6",
    "description": "CloudApplicationStateChangeEvent illustration",
    "domain": "Commercial",
    "eventId": "569",
    "eventTime": "2021-09-27T07:43:59.059Z",
    "eventType": "CloudApplicationStateChangeEvent",
    "priority": "1",
    "timeOcurred": "2021-09-27T07:43:59.059Z",
    "title": "CloudApplicationStateChangeEvent",
    "event": {
      "cloudApplication": {
        "id": "2287",
        "href":
"http://server:port/privateOptimizedBinding/v5/cloudApplication/2287",
        "operationalState": "enable",
        "@type": "CloudApplication",
        "@schemaLocation": "http://server:port/CloudApplication.schema.json",
        "@baseType": "LogicalResource"
      }
    },
    "reportingSystem": {
      "id": "123",
      "name": "CRM app",
      "href": "",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "source": {
      "id": "123",
      "name": "CRM app",
      "href": "",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
```

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```
    },
    "@baseType": "Event",
    "@type": "CloudApplicationStateChangeEvent"
  }
```

User Equipment Specification Attribute Value Change Event

Message example for UserEquipmentSpecificationAttributeValueChangeEvent event

```
Content-Type: application/json

{
    "correlationId": "8a1d7f00-245b",
    "description": "UserEquipmentSpecificationAttributeValueChangeEvent
illustration",
    "domain": "Commercial",
    "eventId": "467c-937d-9afe827e85ee",
    "eventTime": "2022-09-14T12:14:28.774Z",
    "eventType": "UserEquipmentSpecificationAttributeValueChangeEvent",
    "priority": "3",
    "timeOcurred": "2022-09-14T12:14:23.473Z",
    "title": "UserEquipmentSpecificationAttributeValueChangeEvent",
    "event": {
        "userEquipmentSpecification": {
            "href":
"https://host:port/privateOptimizedBinding/v5/UserEquipmentSpecification/9693",
            "id": "9693",
            "@type": "UserEquipmentSpecification",
            "name": "MySpecificationName",
            "isBundle": true
        }
    },
    "reportingSystem": {
        "id": "533",
        "name": "APP-963",
        "href": "",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "200",
        "name": "APP-893",
        "href": "",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "UserEquipmentSpecificationAttributeValueChangeEvent"
}
```

User Equipment Specification Create Event

Message example for UserEquipmentSpecificationCreateEvent event

```
Content-Type: application/json

{
```

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```
    "correlationId": "126-54f",
    "description": "UserEquipmentSpecificationCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "125",
    "eventTime": "2021-09-27T07:43:59.059Z",
    "eventType": "UserEquipmentSpecificationCreateEvent",
    "priority": "1",
    "timeOcurred": "2021-09-27T07:43:59.059Z",
    "title": "UserEquipmentSpecificationCreateEvent",
    "event": {
      "userEquipmentSpecification": {
        "id": "11140",
        "href":
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11140",
        "description": "This is an example of a user equipment specification that we
want for or binded UE.",
        "version": "1.0",
        "isPortable": false,
        "isWearable": false,
        "hasNetworkInterface": true,
        "vendor": "MyUEVendor",
        "model": "MyUEModel",
        "name": "MySpecificationName",
        "isBundle": false,
        "resourceSpecCharacteristic":[
            {
                "name": "MSISDN",
                "valueType": "text",
                "@type": "CharacteristicSpecification"
            }
        ],
        "@type": "UserEquipmentSpecification",
        "@schemaLocation":
"http://server:port/UserEquipmentSpecification.schema.json",
        "@baseType": "PhysicalResourceSpecification"
      }
    },
    "relatedParty": [
      {
        "id": "56d",
        "role": "Issuer",
        "@type": "RelatedParty",
        "@referredType": "Organization"
      }
    ],
    "reportingSystem": {
      "id": "123",
      "name": "CRM app",
      "href": "",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "source": {
      "id": "123",
      "name": "CRM app",
      "href": "",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "UserEquipmentSpecificationCreateEvent"
}
```


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User Equipment Specification Delete Event

This operation deletes a UserEquipmentSpecification entity.

```
Content-Type: application/json

{
    "correlationId": "145d-8hh",
    "description": "UserEquipmentSpecificationDeleteEvent illustration",
    "domain": "Commercial",
    "eventId": "523",
    "eventTime": "2021-09-27T07:43:59.059Z",
    "eventType": "UserEquipmentSpecificationDeleteEvent",
    "priority": "3",
    "timeOcurred": "2021-09-27T07:43:59.059Z",
    "title": "UserEquipmentSpecificationDeleteEvent",
    "event": {
      "userEquipmentSpecification": {
        "id": "9693",
        "href":
"https://host:port/privateOptimizedBinding/v5/userEquipmentSpecification/9693",
        "@type": "UserEquipmenSpecification"
      }
    },
    "reportingSystem": {
      "id": "123",
      "name": "Reporting app",
      "href": "",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "source": {
      "id": "123",
      "name": "Reporting app",
      "href": "",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "UserEquipmentSpecificationDeleteEvent"
}
```

Cloud Application Specification Attribute Value Change Event

Message example for CloudApplicationSpecificationAttributeValueChangeEvent event

```
Content-Type: application/json

{
    "correlationId": "8a1d7f00-245b",
    "description": "CloudApplicationSpecificationAttributeValueChangeEvent
illustration",
    "domain": "Commercial",
    "eventId": "467c-937d-9afe827e85ee",
    "eventTime": "2022-09-14T12:14:28.774Z",
    "eventType": "CloudApplicationSpecificationAttributeValueChangeEvent",
    "priority": "3",
    "timeOcurred": "2022-09-14T12:14:23.473Z",
```

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```
    "title": "CloudApplicationSpecificationAttributeValueChangeEvent",
    "event": {
        "cloudApplicationSpecification": {
            "href":
"https://host:port/privateOptimizedBinding/v5/CloudApplicationSpecification/9693",
            "id": "9693",
            "@type": "CloudApplicationSpecification",
            "name": "MySpecificationName",
            "isBundle": true
        }
    },
    "reportingSystem": {
        "id": "533",
        "name": "APP-963",
        "href": "",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "source": {
        "id": "200",
        "name": "APP-893",
        "href": "",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "CloudApplicationSpecificationAttributeValueChangeEvent"
}
```


Cloud Application Specification Create Event

Message example for CloudApplicationSpecificationCreateEvent event


```
Content-Type: application/json

{
    "correlationId": "126-54f",
    "description": "CloudApplicationSpecificationCreateEvent illustration",
    "domain": "Commercial",
    "eventId": "125",
    "eventTime": "2021-09-27T07:43:59.059Z",
    "eventType": "CloudApplicationSpecificationCreateEvent",
    "priority": "1",
    "timeOcurred": "2021-09-27T07:43:59.059Z",
    "title": "CloudApplicationSpecificationCreateEvent",
    "event": {
      "cloudApplicationSpecification": {
        "description": "This is an example of a cloud application specification.",
        "version": "1.0",
        "name": "MyAppSpecificationName",
        "resourceSpecCharacteristic":[
            {
                "name": "cloudProvider",
                "valueType": "text",
                "@type" : "CharacteristicSpecification"
            },
            {
                "name": "cloudRegion",
                "valueType": "text",
                "@type" : "CharacteristicSpecification"
```


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```
            },
            {
                "name": "cloudReference",
                "valueType": "text",
                "@type" : "CharacteristicSpecification"
            }
        ],
        "@type": "CloudApplicationSpecification",
        "@schemaLocation":
"http://server:port/CloudApplicationSpecification.schema.json",
        "@baseType": "LogicalResourceSpecification"
      }
    },
    "relatedParty": [
      {
        "id": "56d",
        "role": "Issuer",
        "@type": "RelatedParty",
        "@referredType": "Organization"
      }
    ],
    "reportingSystem": {
      "id": "123",
      "name": "CRM app",
      "href": "",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "source": {
      "id": "123",
      "name": "CRM app",
      "href": "",
      "@type": "ReportingResource",
      "@referredType": "LogicalResource"
    },
    "@baseType": "Event",
    "@type": "CloudApplicationSpecificationCreateEvent"
}
```

Cloud Application Specification Delete Event

This operation deletes a CloudApplicationSpecification entity.

```
Content-Type: application/json

{
    "correlationId": "145d-8hh",
    "description": "CloudApplicationSpecificationDeleteEvent illustration",
    "domain": "Commercial",
    "eventId": "523",
    "eventTime": "2021-09-27T07:43:59.059Z",
    "eventType": "CloudApplicationSpecificationDeleteEvent",
    "priority": "3",
    "timeOcurred": "2021-09-27T07:43:59.059Z",
    "title": "CloudApplicationSpecificationDeleteEvent",
    "event": {
      "cloudApplicationSpecification": {
        "id": "9693",
        "href":
"https://host:port/privateOptimizedBinding/v5/cloudApplicationSpecification/9693",
```

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```
          "@type": "CloudApplicationSpecification"
        }
      },
      "reportingSystem": {
        "id": "123",
        "name": "Reporting app",
        "href": "",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
      },
      "source": {
        "id": "123",
        "name": "Reporting app",
        "href": "",
        "@type": "ReportingResource",
        "@referredType": "LogicalResource"
      },
      "@baseType": "Event",
      "@type": "CloudApplicationSpecificationDeleteEvent"
  }
```


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API OPERATIONS

Remember the following Uniform Contract:

Operation on Entities	Uniform API Operation	Description
Query Entities	GET Resource	GET must be used to retrieve a representation of a resource.
Create Entity	POST Resource	POST must be used to create a new resource
Partial Update of an Entity	PATCH Resource	PATCH must be used to partially update a resource
Remove an Entity	DELETE Resource	DELETE must be used to remove a resource
Execute an Action on an Entity	POST on TASK Resource	POST must be used to execute Task Resources
Other Request Methods	POST on TASK Resource	GET and POST must not be used to tunnel other request methods.

Filtering and attribute selection rules are described in the TMF REST Design Guidelines.

Notifications are also described in a subsequent section.

Operations on UserEquipment

List or find UserEquipment objects

`GET /userEquipment?fields=…`

Description

This operation list UserEquipment entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a list of User Equipments. The given criteria is the @type (UserEquipment) and operationalState (enable).

Request

```
GET
/userEquipment?fields=id,href,operationalState,powerState,versionNumber,@type&operat
ionalStateState=enable&versionNumber=1.0
Content-Type: application/json
```

Response

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```
200

[
  {
    "id": "1114",
    "href": "http://server:port/privateOptimizedBinding/v5/userEquipment/1114",
    "operationalState": "enable",
    "versionNumber": "1.0",
    "powerState": "Unknown",
    "description": "This is an example of a user equipment that we want to bind to a
cloud application.",
    "name": "Security Device XYZ",
    "note": [
      {
        "text": "something about this resource",
        "@type": "Note"
      }
    ],
    "place": [{
      "role": "HeadQuarter",
      "place": {
          "id": "2b8cea8f-642d-42e0-bdb2-6f47fabb3f07",
          "href": "https://host/tmf-
api/geographicAddressManagement/v4/geographicAddress/2b8cea8f-642d-42e0-bdb2-
6f47fabb3f07",
          "streetNr": "225",
          "streetNrSuffix": "B",
          "streetNrLast": "",
          "streetNrLastSuffix": "",
          "streetName": " Strathmore",
          "streetType": "Terrace",
          "streetSuffix": "",
          "postcode": "5004",
          "locality": "Brighton.",
          "city": "Brighton",
          "stateOrProvince": "SA",
          "country": "Australia",
          "@type": "GeographicAddress"
      },
      "@type": "RelatedPlaceRefOrValue"
    }],
    "activationFeature": [
      {
        "id": "1",
        "isEnabled": true,
        "name": "privateOptimizedBindingFeature",
        "featureCharacteristic": [
          {
            "name": "privateOptimizedBindingCloudAppHref",
            "value":
"http://server:port/privateOptimizedBinding/v5/cloudApplication/2287",
            "@type": "StringCharacteristic"
          }
        ],
        "@type": "BindingFeature"
      }
    ],
    "resourceCharacteristic": [
      {
        "name": "MSISDN",
        "valueType": "string",
        "value": "3345667788643",
```

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```
        "@type": "StringCharacteristic"
      }
    ],
    "resourceSpecification" : {
      "id" : "11141",
      "href" :
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11141",
      "@type" : "ResourceSpecificationRef"
    },
    "@type": "UserEquipment",
    "@schemaLocation": "http://server:port/UserEquipment.schema.json",
    "@baseType": "PhysicalResource"
  },
  {
    "id": "42",
    "href": "http://server:port/privateOptimizedBinding/v5/userEquipment/42",
    "operationalState": "enable",
    "versionNumber": "1.0",
    "powerState": "Full Power Applied",
    "description": "This is an example of a user equipment that we want to bind to a
cloud application.",
    "name": "Security Device XYZ",
    "note": [
      {
        "text": "something more about this resource",
        "@type": "Note"
      }
    ],
    "place": [{
      "role": "HeadQuarter",
      "place": {
          "id": "2b8cea8f-642d-42e0-bdb2-6f47fabb3f07",
          "href": "https://host/tmf-
api/geographicAddressManagement/v4/geographicAddress/2b8cea8f-642d-42e0-bdb2-
6f47fabb3f07",
          "streetNr": "225",
          "streetNrSuffix": "B",
          "streetNrLast": "",
          "streetNrLastSuffix": "",
          "streetName": " Strathmore",
          "streetType": "Terrace",
          "streetSuffix": "",
          "postcode": "5004",
          "locality": "Brighton.",
          "city": "Brighton",
          "stateOrProvince": "SA",
          "country": "Australia",
          "@type": "GeographicAddress"
      },
      "@type": "RelatedPlaceRefOrValue"
    }],
    "activationFeature": [
      {
        "id": "1",
        "isEnabled": true,
        "name": "privateOptimizedBindingFeature",
        "featureCharacteristic": [
          {
            "name": "privateOptimizedBindingCloudAppHref",
            "value":
"http://server:port/privateOptimizedBinding/v5/cloudApplication/2287",
            "@type": "StringCharacteristic"
          }
```


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```
        ],
        "@type": "BindingFeature"
      }
    ],
    "resourceCharacteristic": [
      {
        "name": "MSISDN",
        "valueType": "string",
        "value": "3345667788644",
        "@type": "StringCharacteristic"
      }
    ],
    "resourceSpecification" : {
      "id" : "11141",
      "href" :
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11141",
      "@type" : "ResourceSpecificationRef"
    },
    "@type": "UserEquipment",
    "@schemaLocation": "http://server:port/UserEquipment.schema.json",
    "@baseType": "PhysicalResource"
  }
]
```


Retrieves a UserEquipment by ID

`GET /userEquipment/{id}?fields=…`

Description

This operation retrieves a UserEquipment entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving an activated UserEquipment with a binding feature. The given criteria is the Resource id 1114.

Request

```
GET /userEquipment/444
Content-Type: application/json
```

Response


```
200

{
    "id": "1114",
    "href": "http://server:port/privateOptimizedBinding/v5/userEquipment/1114",
    "description": "This is an example of a user equipment that we want to bind to a
cloud application.",
    "name": "Security Device XYZ",
    "operationalState": "enable",
```


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```
    "note": [{
            "text": "something about this resource",
            "@type": "Note"
        }],
    "place": [{
        "role": "HeadQuarter",
        "place": {
            "id": "2b8cea8f-642d-42e0-bdb2-6f47fabb3f07",
            "href": "https://host/tmf-
api/geographicAddressManagement/v4/geographicAddress/2b8cea8f-642d-42e0-bdb2-
6f47fabb3f07",
            "@type": "PlaceRef",
            "@referredType" : "GeographicAddress"
        },
        "@type": "RelatedPlaceRefOrValue"
    }],
    "activationFeature": [{
            "id": "1",
            "isEnabled": true,
            "name": "privateOptimizedBindingFeature",
            "featureCharacteristic": [{
                    "name": "privateOptimizedBindingCloudAppHref",
                    "value":
"http://server:port/privateOptimizedBinding/v5/cloudApplication/2287",
                    "@type": "StringCharacteristic"
                }],
            "@type": "BindingFeature"
        }],
    "resourceCharacteristic": [{
            "name": "MSISDN",
            "valueType": "string",
            "value": "3345667788643",
            "@type": "StringCharacteristic"
        }],
    "resourceSpecification" : {
        "id" : "11140",
        "href":
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11140",
        "@type" : "ResourceSpecificationRef"
    },
    "@type": "UserEquipment",
    "@schemaLocation": "http://server:port/UserEquipment.schema.json",
    "@baseType": "PhysicalResource"
}
```


Here's an example of a request for retrieving a UserEquipment without a binding feature. The given criteria is the Resource id 42.

Request

```
GET /userEquipment/42
Content-Type: application/json
```

Response

```
200
```

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```
{
    "id": "42",
    "href": "http://server:port/privateOptimizedBinding/v5/userEquipment/42",
    "description": "This is an example of a user equipment that we want to bind to a
cloud application.",
    "name": "Security Device XYZ",
    "operationalState": "enable",
    "note": [{
        "text": "something about this resource",
        "@type": "Note"
    }],
    "place": [{
        "role": "HeadQuarter",
        "place": {
            "id": "2b8cea8f-642d-42e0-bdb2-6f47fabb3f07",
            "href": "https://host/tmf-
api/geographicAddressManagement/v4/geographicAddress/2b8cea8f-642d-42e0-bdb2-
6f47fabb3f07",
            "streetNr": "225",
            "streetNrSuffix": "B",
            "streetNrLast": "",
            "streetNrLastSuffix": "",
            "streetName": " Strathmore",
            "streetType": "Terrace",
            "streetSuffix": "",
            "postcode": "5004",
            "locality": "Brighton.",
            "city": "Brighton",
            "stateOrProvince": "SA",
            "country": "Australia",
            "@type": "GeographicAddress"
        },
        "@type": "RelatedPlaceRefOrValue"
    }],
    "resourceCharacteristic": [{
        "name": "MSISDN",
        "valueType": "string",
        "value": "3345667788643",
        "@type": "StringCharacteristic"
    }],
    "resourceSpecification" : {
        "id" : "11140",
        "href":
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11140",
        "@type" : "ResourceSpecificationRef"
    },
    "@type": "UserEquipment",
    "@schemaLocation": "http://server:port/UserEquipment.schema.json",
    "@baseType": "PhysicalResource"
}
```


Creates a UserEquipment

`POST /userEquipment?fields=…`

Description

This operation creates a UserEquipment entity.

Mandatory Attributes

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Mandatory Attributes	Rule
operationalState	
@type	

Usage samples

Here's an example of a request and response for creating a new UserEquipment

Request

```
POST /userEquipment
Content-Type: application/json

{
  "description": "This is an example of a user equipment that we want to bind to a
cloud application.",
   "name": "Security Device XYZ",
   "operationalState": "disable",
   "note": [{
       "text": "something about this resource",
       "@type": "Note"
   }],
   "place": [{
           "role": "Area",
           "place": {
              "id": "2b8cea8f-642d-42e0-bdb2-6f47fabb3f07",
              "href": "https://host/tmf-
api/geographicAddressManagement/v4/geographicAddress/2b8cea8f-642d-42e0-bdb2-
6f47fabb3f07",
              "streetNr": "225",
              "streetNrSuffix": "B",
              "streetNrLast": "",
              "streetNrLastSuffix": "",
              "streetName": " Strathmore",
              "streetType": "Terrace",
              "streetSuffix": "",
              "postcode": "5004",
              "locality": "Brighton.",
              "city": "Brighton",
              "stateOrProvince": "SA",
              "country": "Australia",
              "@type": "GeographicAddress"
           },
           "@type": "RelatedPlaceRefOrValue"
   }],
   "resourceCharacteristic": [{
       "name": "MSISDN",
       "valueType": "string",
       "value": "3345667788643",
       "@type": "StringCharacteristic"
   }],
   "resourceSpecification" : {
    "id" : "11140",
    "href":
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11140",
    "@type" : "ResourceSpecificationRef"
   },
   "@type": "UserEquipment",
```

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```
    "@schemaLocation": "http://server:port/UserEquipment.schema.json",
    "@baseType": "PhysicalResource"
  }
```

Response

```
201

{
    "id": "42",
    "href": "http://server:port/privateOptimizedBinding/v5/userEquipment/42",
    "description": "This is an example of a user equipment that we want to bind to a
cloud application.",
    "name": "Security Device XYZ",
    "operationalState": "disable",
    "note": [{
        "text": "something about this resource",
        "@type": "Note"
    }],
    "place": [{
        "role": "Area",
        "place": {
            "id": "2b8cea8f-642d-42e0-bdb2-6f47fabb3f07",
            "href": "https://host/tmf-
api/geographicAddressManagement/v4/geographicAddress/2b8cea8f-642d-42e0-bdb2-
6f47fabb3f07",
            "streetNr": "225",
            "streetNrSuffix": "B",
            "streetNrLast": "",
            "streetNrLastSuffix": "",
            "streetName": " Strathmore",
            "streetType": "Terrace",
            "streetSuffix": "",
            "postcode": "5004",
            "locality": "Brighton.",
            "city": "Brighton",
            "stateOrProvince": "SA",
            "country": "Australia",
            "@type": "GeographicAddress"
        },
        "@type" : "RelatedPlaceRefOrValue"
    }],
    "resourceCharacteristic": [{
        "name": "MSISDN",
        "valueType": "string",
        "value": "3345667788643",
        "@type": "StringCharacteristic"
    }],
    "resourceSpecification" : {
        "id": "11140",
        "href":
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11140",
        "@type" : "ResourceSpecificationRef"
    },
    "@type": "UserEquipment",
    "@schemaLocation": "http://server:port/UserEquipment.schema.json",
    "@baseType": "PhysicalResource"
}
```

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Updates partially a UserEquipment

PATCH /userEquipment/{id}?fields=…

Description

This operation allows partial updates of a UserEquipment entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optiona. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
administrativeState	
href	
id	
manufactureDate	
serialNumber	
versionNumber	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
activationFeature	
attachment	
category	
description	
endOperatingDate	
externalIdentifier	
intent	
name	
note	
operationalState	
place	
powerState	
relatedParty	

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Patchable Attributes	Rule
resourceCharacteristic	
resourceOrderItem	
resourceRelationship	
resourceSpecification	
resourceStatus	
resourceVersion	
startOperatingDate	
usageState	
validFor	

Usage samples

Update a UserEquipment using json

Request

```
PATCH /userEquipment/{id}
Content-Type: application/json

{

    "operationalState": "enable",
    "note": [{
        "text": "This is an updated note",
        "@type": "Note"
    }]
}
```

Response

```
200

{
    "id": "1114",
    "href": "http://server:port/privateOptimizedBinding/v5/userEquipment/1114",
    "description": "This is an example of a user equipment that we want to bind to a
cloud application.",
    "name": "Security Device XYZ",
    "operationalState": "enable",
    "note": [{
      "text": "This is an updated note",
      "@type": "Note"
      }],
    "place": [{
      "role": "HeadQuarter",
      "place": {
            "id": "2b8cea8f-642d-42e0-bdb2-6f47fabb3f07",
            "href": "https://host/tmf-
api/geographicAddressManagement/v4/geographicAddress/2b8cea8f-642d-42e0-bdb2-
```

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```
6f47fabb3f07",
            "streetNr": "225",
            "streetNrSuffix": "B",
            "streetNrLast": "",
            "streetNrLastSuffix": "",
            "streetName": " Strathmore",
            "streetType": "Terrace",
            "streetSuffix": "",
            "postcode": "5004",
            "locality": "Brighton.",
            "city": "Brighton",
            "stateOrProvince": "SA",
            "country": "Australia",
            "@type": "GeographicAddress"
        },
        "@type": "RelatedPlaceRefOrValue"
    }],
    "resourceCharacteristic": [{
      "name": "MSISDN",
      "valueType": "string",
      "value": "3345667788643",
      "@type": "StringCharacteristic"
  }],
  "resourceSpecification" : {
    "id" : "11140",
    "href":
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11140",
    "@type" : "ResourceSpecificationRef"
    },
  "@type": "UserEquipment",
  "@schemaLocation": "http://server:port/UserEquipment.schema.json",
  "@baseType": "PhysicalResource"
}
```

Update a UserEquipment using json-patch with query parameters

Request

```
PATCH
/userEquipment/{id}?fields=id,href,operationalState,name,description,note,@type
Content-Type: application/json-patch+json

[
    {   "op":"replace",
        "path": "/operationalState",
        "value": "enable"
    },
    {   "op":"replace",
        "path" : "/note/1",
        "value" :{
            "text": "This is an updated note"
        }
    }
]
```

Response

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```
200

{
    "id": "1114",
    "href": "http://server:port/privateOptimizedBinding/v5/userEquipment/1114",
    "description": "This is an example of a user equipment that we want to bind to a
cloud application.",
    "name": "Security Device XYZ",
    "operationalState": "enable",
    "note": [{
      "text": "This is an updated note",
      "@type": "Note"
    }],
    "@type": "UserEquipment"
}
```

Update a UserEquipment using json-patch with No query parameters

Request

```
PATCH /userEquipment/{id}
Content-Type: application/json-patch+json

[
    {   "op":"replace",
        "path": "/operationalState",
        "value": "enable"
    },
    {   "op":"replace",
        "path" : "/note/1",
        "value" :{
            "text": "This is an updated note"
        }
    }
]
```

Response

```
200

{
    "id": "1114",
    "href": "http://server:port/privateOptimizedBinding/v5/userEquipment/1114",
    "description": "This is an example of a user equipment that we want to bind to a
cloud application.",
    "name": "Security Device XYZ",
    "operationalState": "enable",
    "note": [{
      "text": "This is an updated note",
      "@type": "Note"
      }],
    "place": [{
      "role": "HeadQuarter",
      "place": {
            "id": "2b8cea8f-642d-42e0-bdb2-6f47fabb3f07",
            "href": "https://host/tmf-
```

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```
api/geographicAddressManagement/v4/geographicAddress/2b8cea8f-642d-42e0-bdb2-
6f47fabb3f07",
            "streetNr": "225",
            "streetNrSuffix": "B",
            "streetNrLast": "",
            "streetNrLastSuffix": "",
            "streetName": " Strathmore",
            "streetType": "Terrace",
            "streetSuffix": "",
            "postcode": "5004",
            "locality": "Brighton.",
            "city": "Brighton",
            "stateOrProvince": "SA",
            "country": "Australia",
            "@type": "GeographicAddress"
        },
        "@type": "RelatedPlaceRefOrValue"
    }],
    "resourceCharacteristic": [{
      "name": "MSISDN",
      "valueType": "string",
      "value": "3345667788643",
      "@type": "StringCharacteristic"
  }],
  "resourceSpecification" : {
    "id" : "11140",
    "href":
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11140",
    "@type" : "ResourceSpecificationRef"
    },
  "@type": "UserEquipment",
  "@schemaLocation": "http://server:port/UserEquipment.schema.json",
  "@baseType": "PhysicalResource"
}
```

Update a UserEquipment using json-patch-query

Request

```
PATCH /userEquipment/{id}
Content-Type: application/json-patch-query+json

[
    {   "op":"replace",
        "path": "/operationalState?name=Security Device XYZ",
        "value": "enable"
    },
    {   "op":"replace",
        "path" : "/note/1?name=Security Device XYZ",
        "value" :{
            "text": "This is an updated note"
        }
    }
]
```

Response

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```
200

{
    "id": "1114",
    "href": "http://server:port/privateOptimizedBinding/v5/userEquipment/1114",
    "description": "This is an example of a user equipment that we want to bind to a
cloud application.",
    "name": "Security Device XYZ",
    "operationalState": "enable",
    "note": [{
      "text": "This is an updated note",
      "@type": "Note"
      }],
    "place": [{
      "role": "HeadQuarter",
      "place": {
            "id": "2b8cea8f-642d-42e0-bdb2-6f47fabb3f07",
            "href": "https://host/tmf-
api/geographicAddressManagement/v4/geographicAddress/2b8cea8f-642d-42e0-bdb2-
6f47fabb3f07",
            "streetNr": "225",
            "streetNrSuffix": "B",
            "streetNrLast": "",
            "streetNrLastSuffix": "",
            "streetName": " Strathmore",
            "streetType": "Terrace",
            "streetSuffix": "",
            "postcode": "5004",
            "locality": "Brighton.",
            "city": "Brighton",
            "stateOrProvince": "SA",
            "country": "Australia",
            "@type": "GeographicAddress"
        },
        "@type": "RelatedPlaceRefOrValue"
    }],
    "resourceCharacteristic": [{
      "name": "MSISDN",
      "valueType": "string",
      "value": "3345667788643",
      "@type": "StringCharacteristic"
  }],
  "resourceSpecification" : {
    "id" : "11140",
    "href":
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11140",
    "@type" : "ResourceSpecificationRef"
    },
  "@type": "UserEquipment",
  "@schemaLocation": "http://server:port/UserEquipment.schema.json",
  "@baseType": "PhysicalResource"
}
```

Update a UserEquipment using merge-patch

Request

```
PATCH /userEquipment/{id}
Content-Type: application/merge-patch+json
```

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```
{

    "operationalState": "enable",
    "note": [{
        "text": "This is an updated note",
        "@type": "Note"
    }]
}
```

Response

```
200

{
    "id": "1114",
    "href": "http://server:port/privateOptimizedBinding/v5/userEquipment/1114",
    "description": "This is an example of a user equipment that we want to bind to a
cloud application.",
    "name": "Security Device XYZ",
    "operationalState": "enable",
    "note": [{
      "text": "This is an updated note",
      "@type": "Note"
      }],
    "place": [{
      "role": "HeadQuarter",
      "place": {
            "id": "2b8cea8f-642d-42e0-bdb2-6f47fabb3f07",
            "href": "https://host/tmf-
api/geographicAddressManagement/v4/geographicAddress/2b8cea8f-642d-42e0-bdb2-
6f47fabb3f07",
            "streetNr": "225",
            "streetNrSuffix": "B",
            "streetNrLast": "",
            "streetNrLastSuffix": "",
            "streetName": " Strathmore",
            "streetType": "Terrace",
            "streetSuffix": "",
            "postcode": "5004",
            "locality": "Brighton.",
            "city": "Brighton",
            "stateOrProvince": "SA",
            "country": "Australia",
            "@type": "GeographicAddress"
        },
        "@type": "RelatedPlaceRefOrValue"
    }],
    "resourceCharacteristic": [{
      "name": "MSISDN",
      "valueType": "string",
      "value": "3345667788643",
      "@type": "StringCharacteristic"
  }],
  "resourceSpecification" : {
    "id" : "11140",
    "href":
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11140",
    "@type" : "ResourceSpecificationRef"
    },
```

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```
  "@type": "UserEquipment",
  "@schemaLocation": "http://server:port/UserEquipment.schema.json",
  "@baseType": "PhysicalResource"
}
```

Update(merge-patch) a UserEquipment to add the bindingFeature

Request

```
PATCH /userEquipment/{id}
Content-Type: application/merge-patch+json

{

    "activationFeature": [{
        "id": "1",
        "isEnabled": false,
        "name": "privateOptimizedBindingFeature",
        "featureCharacteristic": [{
                "name": "privateOptimizedBindingCloudAppHref",
                "value":
"http://server:port/privateOptimizedBinding/v5/cloudApplication/2287",
                "@type" : "StringCharacteristic"
            }],
        "@type": "BindingFeature"
    },{
            "id": "2",
            "isEnabled": true,
            "name": "randomFeature",
            "@type": "Feature",
            "featureCharacteristic": [{
                    "name": "testCharecteristic",
                    "value": "testValue",
                    "@type": "StringCharacteristic"
                }]
        }]
}
```

Response

```
200

{
    "id": "1114",
    "href": "http://server:port/privateOptimizedBinding/v5/userEquipment/1114",
    "description": "This is an example of a user equipment that we want to bind to a
cloud application.",
    "name": "Security Device XYZ",
    "operationalState": "enable",
    "note": [{
        "text": "This is an updated note",
        "@type": "Note"
    }],
    "place": [{
        "role": "HeadQuarter",
        "place": {
            "id": "2b8cea8f-642d-42e0-bdb2-6f47fabb3f07",
```

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```
            "href": "https://host/tmf-
api/geographicAddressManagement/v4/geographicAddress/2b8cea8f-642d-42e0-bdb2-
6f47fabb3f07",
            "streetNr": "225",
            "streetNrSuffix": "B",
            "streetNrLast": "",
            "streetNrLastSuffix": "",
            "streetName": " Strathmore",
            "streetType": "Terrace",
            "streetSuffix": "",
            "postcode": "5004",
            "locality": "Brighton.",
            "city": "Brighton",
            "stateOrProvince": "SA",
            "country": "Australia",
            "@type": "GeographicAddress"
        },
        "@type": "RelatedPlaceRefOrValue"
    }],
    "activationFeature": [{
        "id": "1",
        "isEnabled": false,
        "name": "privateOptimizedBindingFeature",
        "featureCharacteristic": [{
            "name": "privateOptimizedBindingCloudAppHref",
            "value":
"http://server:port/privateOptimizedBinding/v5/cloudApplication/2287",
            "@type": "StringCharacteristic"
        }],
        "@type": "BindingFeature"
        },{
            "id": "2",
            "isEnabled": true,
            "name": "randomFeature",
            "@type": "Feature",
            "featureCharacteristic": [{
                "name": "privateOptimizedBindingCloudAppHref",
                "value":
"http://server:port/privateOptimizedBinding/v5/cloudApplication/2287",
                "@type": "StringCharacteristic"
            }]
        }],
    "resourceCharacteristic": [{
        "name": "MSISDN",
        "valueType": "string",
        "value": "3345667788643",
        "@type": "StringCharacteristic"
    }],
    "resourceSpecification" : {
        "id" : "11143",
        "href":
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11143",
        "@type": "ResourceSpecificationRef"
    },
    "@type": "UserEquipment",
    "@schemaLocation": "http://server:port/UserEquipment.schema.json",
    "@baseType": "PhysicalResource"
}
```


Update(merge-patch) a UserEquipment to activate the existing bindingFeature

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Request

```
PATCH /userEquipment/{id}
Content-Type: application/merge-patch+json

{

    "activationFeature": [{
        "id": "1",
        "isEnabled": true,
        "name": "privateOptimizedBindingFeature",
        "featureCharacteristic": [{
                "name": "privateOptimizedBindingCloudAppHref",
                "value":
"http://server:port/privateOptimizedBinding/v5/cloudApplication/2287",
                "@type": "StringCharacteristic"
            }],
        "@type": "BindingFeature"
        },{
            "id": "2",
            "isEnabled": true,
            "name": "randomFeature",
            "@type": "Feature",
            "featureCharacteristic": [{
                    "name": "testCharecteristic",
                    "value": "testValue",
                    "@type": "StringCharacteristic"
                }]
        }]
}
```

Response

```
200

{
    "id": "1114",
    "href": "http://server:port/privateOptimizedBinding/v5/userEquipment/1114",
    "description": "This is an example of a user equipment that we want to bind to a
cloud application.",
    "name": "Security Device XYZ",
    "operationalState": "enable",
    "note": [{
        "text": "This is an updated note",
        "@type": "Note"
    }],
    "place": [{
        "role": "HeadQuarter",
        "place": {
            "id": "2b8cea8f-642d-42e0-bdb2-6f47fabb3f07",
            "href": "https://host/tmf-
api/geographicAddressManagement/v4/geographicAddress/2b8cea8f-642d-42e0-bdb2-
6f47fabb3f07",
            "streetNr": "225",
            "streetNrSuffix": "B",
            "streetNrLast": "",
            "streetNrLastSuffix": "",
            "streetName": " Strathmore",
            "streetType": "Terrace",
```

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```
            "streetSuffix": "",
            "postcode": "5004",
            "locality": "Brighton.",
            "city": "Brighton",
            "stateOrProvince": "SA",
            "country": "Australia",
            "@type": "GeographicAddress"
        },
        "@type": "RelatedPlaceRefOrValue"
    }],
    "activationFeature": [{
        "id": "1",
        "isEnabled": true,
        "name": "privateOptimizedBindingFeature",
        "featureCharacteristic": [{
                "name": "privateOptimizedBindingCloudAppHref",
                "value":
"http://server:port/privateOptimizedBinding/v5/cloudApplication/2287",
                "@type" : "StringCharacteristic"
            }],
        "@type": "BindingFeature"
    },{
            "id": "2",
            "isEnabled": true,
            "name": "randomFeature",
            "@type": "Feature",
            "featureCharacteristic": [{
                    "name": "testCharecteristic",
                    "value": "testValue",
                    "@type": "StringCharacteristic"
                }]
        }],
    "resourceCharacteristic": [{
        "name": "MSISDN",
        "valueType": "string",
        "value": "3345667788643",
        "@type": "StringCharacteristic"
    }],
    "resourceSpecification" : {
        "id" : "11143",
        "href":
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11143",
        "@type" : "ResourceSpecificationRef"
    },
    "@type": "UserEquipment",
    "@schemaLocation": "http://server:port/UserEquipment.schema.json",
    "@baseType": "PhysicalResource"
}
```

Update(merge-patch) a UserEquipment to disable the previously activated bindingFeature

Request

```
PATCH /userEquipment/{id}
Content-Type: application/merge-patch+json

{

    "activationFeature": [{
        "id": "1",
```

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```
        "isEnabled": false,
        "name": "privateOptimizedBindingFeature",
        "featureCharacteristic": [{
                "name": "privateOptimizedBindingCloudAppHref",
                "value":
"http://server:port/privateOptimizedBinding/v5/cloudApplication/2287",
                "@type": "StringCharacteristic"
            }],
        "@type": "BindingFeature"
    },{
            "id": "2",
            "isEnabled": true,
            "name": "randomFeature",
            "@type": "Feature",
            "featureCharacteristic": [{
                    "name": "testCharecteristic",
                    "value": "testValue",
                    "@type": "StringCharacteristic"
                }]
        }]
}
```


Response


```
200

{
    "id": "1114",
    "href": "http://server:port/privateOptimizedBinding/v5/userEquipment/1114",
    "description": "This is an example of a user equipment that we want to bind to a
cloud application.",
    "name": "Security Device XYZ",
    "operationalState": "enable",
    "note": [{
        "text": "This is an updated note",
        "@type": "Note"
    }],
    "place": [{
        "role": "HeadQuarter",
        "place": {
            "id": "2b8cea8f-642d-42e0-bdb2-6f47fabb3f07",
            "href": "https://host/tmf-
api/geographicAddressManagement/v4/geographicAddress/2b8cea8f-642d-42e0-bdb2-
6f47fabb3f07",
            "streetNr": "225",
            "streetNrSuffix": "B",
            "streetNrLast": "",
            "streetNrLastSuffix": "",
            "streetName": " Strathmore",
            "streetType": "Terrace",
            "streetSuffix": "",
            "postcode": "5004",
            "locality": "Brighton.",
            "city": "Brighton",
            "stateOrProvince": "SA",
            "country": "Australia",
            "@type": "GeographicAddress"
        },
        "@type": "RelatedPlaceRefOrValue"
    }],
```


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```
    "activationFeature": [{
        "id": "1",
        "isEnabled": false,
        "name": "privateOptimizedBindingFeature",
        "featureCharacteristic": [{
            "name": "privateOptimizedBindingCloudAppHref",
            "value":
"http://server:port/privateOptimizedBinding/v5/cloudApplication/2287",
            "@type": "StringCharacteristic"
        }],
        "@type": "BindingFeature"
    },{
            "id": "2",
            "isEnabled": true,
            "name": "randomFeature",
            "@type": "Feature",
            "featureCharacteristic": [{
                "name": "testCharecteristic",
                "value": "testValue",
                "@type": "StringCharacteristic"
            }]
        }],
    "resourceCharacteristic": [{
        "name": "MSISDN",
        "valueType": "string",
        "value": "3345667788643",
        "@type": "StringCharacteristic"
    }],
    "resourceSpecification" : {
        "id": "11143",
        "href":
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11143",
        "@type": "ResourceSpecificationRef"
    },
    "@type": "UserEquipment",
    "@schemaLocation": "http://server:port/UserEquipment.schema.json",
    "@baseType": "PhysicalResource"
}
```

Deletes a UserEquipment

`DELETE /userEquipment/{id}`

Description

This operation deletes a UserEquipment entity.

Usage samples

Here's an example for a request for deleteing specific UserEquipment.

Request

```
DELETE /userEquipment/42
Content-Type: application/json
```

Response

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```
204
```

Operations on CloudApplication

List or find CloudApplication objects

`GET /cloudApplication?fields=…`

Description

This operation list CloudApplication entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a CloudApplications. The given criteria is the @type (CloudApplication) and operationalState (enable).

Request

```
GET
/cloudApplication?fields=id,href,operationalState,value,description,@type&operationa
lState=enable&value=myAppValue
Content-Type: application/json
```

Response

```
200

[
    {
        "id": "2287",
        "href":
"http://server:port/privateOptimizedBinding/v5/cloudApplication/2287",
        "operationalState": "enable",
        "name": "Security Application ABC",
        "description": "this is a dummy description for first cloudApplication",
        "note": [
            {
                "text": "this is a security application supporting TMF759 binding",
                "@type": "Note"
            }
        ],
        "resourceCharacteristic": [
            {
                "name": "cloudProvider",
                "valueType": "string",
                "value": "MS",
                "@type": "StringCharacteristic"
            },
            {
                "name": "cloudRegion",
```

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```
                "valueType": "string",
                "value": "eastus",
                "@type": "StringCharacteristic"
            },
            {
                "name": "cloudReference",
                "valueType": "string",
                "value": "/subscriptions/00000000-0000-0000-0000-
000000000000/resourceGroups/vnet-
rg/providers/Microsoft.Compute/virtualMachines/vmX",
                "@type": "StringCharacteristic"
            }
        ],
        "resourceSpecification" : {
            "id" : "11140",
            "href" :
"http://server:port/privateOptimizedBinding/v5/cloudApplicationSpecification/11140",
            "@type" : "ResourceSpecificationRef"
        },
        "@type": "CloudApplication",
        "@schemaLocation": "http://server:port/CloudApplication.schema.json",
        "@baseType": "LogicalResource"
    },
    {
        "id": "3380",
        "href":
"http://server:port/privateOptimizedBinding/v5/cloudApplication/2287",
        "operationalState": "enable",
        "name": "Security Application ABC",
        "description": "this is a dummy description for second cloudApplication
(game)",
        "note": [
            {
                "text": "this is a security application supporting TMF759 binding",
                "@type": "Note"
            }
        ],
        "resourceCharacteristic": [
            {
                "name": "cloudProvider",
                "valueType": "string",
                "value": "MS2",
                "@type": "StringCharacteristic"
            },
            {
                "name": "cloudRegion",
                "valueType": "string",
                "value": "eastus2",
                "@type": "StringCharacteristic"
            },
            {
                "name": "cloudReference",
                "valueType": "string",
                "value": "/subscriptions2/00000000-0000-0000-0000-
000000000000/resourceGroups/vnet-
rg/providers/Microsoft.Compute/virtualMachines/vmX",
                "@type": "StringCharacteristic"
            },
            {
                "name": "myGameCharacteristic",
                "valueType": "string",
                "value": "My random characteristic",
                "@type": "StringCharacteristic"
```


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```
            }
        ],
        "resourceSpecification" : {
            "id" : "11141",
            "href" :
"http://server:port/privateOptimizedBinding/v5/cloudApplicationSpecification/11141",
            "@type": "ResourceSpecificationRef"
        },
        "@type": "CloudApplication",
        "@schemaLocation": "http://server:port/CloudApplication.schema.json",
        "@baseType": "LogicalResource"
    }
]
```

Retrieves a CloudApplication by ID

`GET /cloudApplication/{id}?fields=…`

Description

This operation retrieves a CloudApplication entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a CloudApplication. The given criteria is the Resource id 2287.

Request

```
GET /cloudApplication/2287
Content-Type: application/json
```

Response

```
200

{
    "id": "2287",
    "href": "http://server:port/privateOptimizedBinding/v5/cloudApplication/2287",
    "operationalState": "enable",
    "name": "Security Application ABC",
    "note": [{
            "text": "this is a security application supporting TMF759 binding",
            "@type": "Note"
        }],
    "resourceCharacteristic": [{
        "name": "cloudProvider",
        "valueType": "string",
        "value": "MS",
        "@type": "StringCharacteristic"
    },
    {
        "name": "cloudRegion",
```

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```
        "valueType": "string",
        "value": "eastus",
        "@type": "StringCharacteristic"
    },
    {
        "name": "cloudReference",
        "valueType": "string",
        "value": "/subscriptions/00000000-0000-0000-0000-
000000000000/resourceGroups/vnet-
rg/providers/Microsoft.Compute/virtualMachines/vmX",
        "@type": "StringCharacteristic"
    }],
    "resourceSpecification" : {
        "id": "11140",
        "href" :
"http://server:port/privateOptimizedBinding/v5/cloudApplicationSpecification/11140",
        "@type" : "ResourceSpecificationRef"
    },
    "@type": "CloudApplication",
    "@schemaLocation": "http://server:port/CloudApplication.schema.json",
    "@baseType": "LogicalResource"
}
```

Creates a CloudApplication

POST /cloudApplication?fields=…

Description

This operation creates a CloudApplication entity.

Mandatory Attributes

Mandatory Attributes	Rule
operationalState	
@type	

Usage samples

Here's an example of a request and response for creating a new CloudApplication

Request

```
POST /cloudApplication
Content-Type: application/json

{

    "operationalState": "enable",
    "name": "Security Application ABC",
    "note": [{
            "text": "this is a security application supporting TMF759 binding",
            "@type": "Note"
        }],
    "resourceCharacteristic": [{
```

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```
            "name": "cloudProvider",
            "valueType": "string",
            "value": "MS",
            "@type": "StringCharacteristic"
        },
        {
            "name": "cloudRegion",
            "valueType": "string",
            "value": "eastus",
            "@type": "StringCharacteristic"
        },
        {
            "name": "cloudReference",
            "valueType": "string",
            "value": "/subscriptions/00000000-0000-0000-0000-
000000000000/resourceGroups/vnet-
rg/providers/Microsoft.Compute/virtualMachines/vmX",
            "@type": "StringCharacteristic"
        }],
    "resourceSpecification" : {
        "id" : "11140",
        "href":
"http://server:port/privateOptimizedBinding/v5/cloudApplicationSpecification/11140",
        "@type" : "ResourceSpecificationRef"
    },
    "@type": "CloudApplication",
    "@schemaLocation": "http://server:port/CloudApplication.schema.json",
    "@baseType": "LogicalResource"
}
```


Response


```
201

{
    "id": "2287",
    "href": "http://server:port/privateOptimizedBinding/v5/cloudApplication/2287",
    "operationalState": "enable",
    "name": "Security Application ABC",
    "note": [{
            "text": "this is a security application supporting TMF759 binding",
            "@type": "Note"
        }],
    "resourceCharacteristic": [{
        "name": "cloudProvider",
        "valueType": "string",
        "value": "MS",
        "@type": "StringCharacteristic"
    },
    {
        "name": "cloudRegion",
        "valueType": "string",
        "value": "eastus",
        "@type": "StringCharacteristic"
    },
    {
        "name": "cloudReference",
        "valueType": "string",
        "value": "/subscriptions/00000000-0000-0000-0000-
000000000000/resourceGroups/vnet-
```


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```
rg/providers/Microsoft.Compute/virtualMachines/vmX",
        "@type": "StringCharacteristic"
    }],
    "resourceSpecification" : {
        "id" : "11140",
        "href" :
"http://server:port/privateOptimizedBinding/v5/cloudApplicationSpecification/11140",
        "@type": "ResourceSpecificationRef"
    },
    "@type": "CloudApplication",
    "@schemaLocation": "http://server:port/CloudApplication.schema.json",
    "@baseType": "LogicalResource"
}
```

Updates partially a CloudApplication

`PATCH /cloudApplication/{id}?fields=…`

Description

This operation allows partial updates of a CloudApplication entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optiona. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
href	
id	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
activationFeature	
administrativeState	
attachment	
category	
description	
endOperatingDate	
externalIdentifier	
intent	
name	

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Patchable Attributes	Rule
note	
operationalState	
place	
relatedParty	
resourceCharacteristic	
resourceOrderItem	
resourceRelationship	
resourceSpecification	
resourceStatus	
resourceVersion	
startOperatingDate	
usageState	
validFor	
value	

Usage samples

Update a CloudApplication using json

Request

```
PATCH /cloudApplication/{id}
Content-Type: application/json

{
    "operationalState": "enable",
    "note": [
        {
            "text": "This is an updated note",
            "@type": "Note"
        }
    ],
    "@type": "CloudApplication"
}
```

Response

```
200

{
    "id": "2287",
    "href": "http://server:port/privateOptimizedBinding/v5/cloudApplication/2287",
    "operationalState": "enable",
    "name": "Security Application ABC",
    "note": [{
```

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```
            "text": "This is an updated note",
            "@type": "Note"
        }],
    "resourceCharacteristic": [{
            "name": "cloudProvider",
            "valueType": "string",
            "value": "MS",
            "@type": "StringCharacteristic"
        },
        {
            "name": "cloudRegion",
            "valueType": "string",
            "value": "eastus",
            "@type": "StringCharacteristic"
        },
        {
            "name": "cloudReference",
            "valueType": "string",
            "value": "/subscriptions/00000000-0000-0000-0000-
000000000000/resourceGroups/vnet-
rg/providers/Microsoft.Compute/virtualMachines/vmX",
            "@type": "StringCharacteristic"
        }],
    "resourceSpecification" : {
        "id" : "11140",

"href":"http://server:port/privateOptimizedBinding/v5/cloudApplicationSpecification/
11140",
        "@type" : "ResourceSpecificationRef"
    },
    "@type": "CloudApplication",
    "@schemaLocation": "http://server:port/CloudApplication.schema.json",
    "@baseType": "LogicalResource"
    }
```

Update a CloudApplication using json-patch with query parameters

Request

```
PATCH /cloudApplication/{id}?fields=id,href,operationalState,name,note,@type
Content-Type: application/json-patch+json

[
    {   "op":"replace",
        "path": "/operationalState",
        "value": "enable"
    },
    {   "op":"replace",
        "path" : "/note/1",
        "value" :{
            "text": "This is an updated note"
        }
    }
]
```

Response

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```
200

{
    "id": "2287",
    "href": "http://server:port/privateOptimizedBinding/v5/cloudApplication/2287",
    "operationalState": "enable",
    "name": "Security Application ABC",
    "note": [{
            "text": "This is an updated note",
            "@type": "Note"
        }],
    "@type": "CloudApplication"
    }
```

Update a CloudApplication using json-patch but no query parameters

Request

```
PATCH /cloudApplication/{id}
Content-Type: application/json-patch+json

[
    {   "op":"replace",
        "path": "/operationalState",
        "value": "enable"
    },
    {   "op":"replace",
        "path" : "/note/1",
        "value" :{
            "text": "This is an updated note"
        }
    }
]
```

Response

```
200

{
    "id": "2287",
    "href": "http://server:port/privateOptimizedBinding/v5/cloudApplication/2287",
    "operationalState": "enable",
    "name": "Security Application ABC",
    "note": [{
            "text": "This is an updated note",
            "@type": "Note"
        }],
    "resourceCharacteristic": [{
            "name": "cloudProvider",
            "valueType": "string",
            "value": "MS",
            "@type": "StringCharacteristic"
        },
        {
            "name": "cloudRegion",
            "valueType": "string",
```

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```
            "value": "eastus",
            "@type": "StringCharacteristic"
        },
        {
            "name": "cloudReference",
            "valueType": "string",
            "value": "/subscriptions/00000000-0000-0000-0000-
000000000000/resourceGroups/vnet-
rg/providers/Microsoft.Compute/virtualMachines/vmX",
            "@type": "StringCharacteristic"
        }],
    "resourceSpecification" : {
        "id" : "11140",

"href":"http://server:port/privateOptimizedBinding/v5/cloudApplicationSpecification/
11140",
        "@type" : "ResourceSpecificationRef"
    },
    "@type": "CloudApplication",
    "@schemaLocation": "http://server:port/CloudApplication.schema.json",
    "@baseType": "LogicalResource"
    }
```

Update a CloudApplication using json-patch-query

Request

```
PATCH /cloudApplication/{id}
Content-Type: application/json-patch-query+json

[
    {   "op":"replace",
        "path": "/operationalState?resourceCharacteristic[0].name=cloudProvider",
        "value": "enable"
    },
    {   "op":"replace",
        "path" : "/note/1?resourceCharacteristic[0].name=cloudProvider",
        "value" :{
            "text": "This is an updated note"
        }
    }
]
```

Response

```
200

{
    "id": "2287",
    "href": "http://server:port/privateOptimizedBinding/v5/cloudApplication/2287",
    "operationalState": "enable",
    "name": "Security Application ABC",
    "note": [{
            "text": "This is an updated note",
            "@type": "Note"
        }],
    "resourceCharacteristic": [{
```

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```
            "name": "cloudProvider",
            "valueType": "string",
            "value": "MS",
            "@type": "StringCharacteristic"
        },
        {
            "name": "cloudRegion",
            "valueType": "string",
            "value": "eastus",
            "@type": "StringCharacteristic"
        },
        {
            "name": "cloudReference",
            "valueType": "string",
            "value": "/subscriptions/00000000-0000-0000-0000-
000000000000/resourceGroups/vnet-
rg/providers/Microsoft.Compute/virtualMachines/vmX",
            "@type": "StringCharacteristic"
        }],
    "resourceSpecification" : {
        "id" : "11140",

"href":"http://server:port/privateOptimizedBinding/v5/cloudApplicationSpecification/
11140",
        "@type" : "ResourceSpecificationRef"
    },
    "@type": "CloudApplication",
    "@schemaLocation": "http://server:port/CloudApplication.schema.json",
    "@baseType": "LogicalResource"
    }
```

Update a CloudApplication using merge-patch

Request

```
PATCH /cloudApplication/{id}
Content-Type: application/merge-patch+json

{
    "operationalState": "enable",
    "note": [
        {
            "text": "This is an updated note",
            "@type": "Note"
        }
    ],
    "@type": "CloudApplication"
}
```

Response

```
200

{
    "id": "2287",
    "href": "http://server:port/privateOptimizedBinding/v5/cloudApplication/2287",
    "operationalState": "enable",
```

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```
    "name": "Security Application ABC",
    "note": [{
            "text": "This is an updated note",
            "@type": "Note"
        }],
    "resourceCharacteristic": [{
            "name": "cloudProvider",
            "valueType": "string",
            "value": "MS",
            "@type": "StringCharacteristic"
        },
        {
            "name": "cloudRegion",
            "valueType": "string",
            "value": "eastus",
            "@type": "StringCharacteristic"
        },
        {
            "name": "cloudReference",
            "valueType": "string",
            "value": "/subscriptions/00000000-0000-0000-0000-
000000000000/resourceGroups/vnet-
rg/providers/Microsoft.Compute/virtualMachines/vmX",
            "@type": "StringCharacteristic"
        }],
    "resourceSpecification" : {
        "id" : "11140",

"href":"http://server:port/privateOptimizedBinding/v5/cloudApplicationSpecification/
11140",
        "@type" : "ResourceSpecificationRef"
    },
    "@type": "CloudApplication",
    "@schemaLocation": "http://server:port/CloudApplication.schema.json",
    "@baseType": "LogicalResource"
    }
```


Deletes a CloudApplication

`DELETE /cloudApplication/{id}`

Description

This operation deletes a CloudApplication entity.

Usage samples

Here's an example for a request for deleteing specific CloudApplication.

Request

```
DELETE /cloudApplication/2287
Content-Type: application/json
```

Response

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```
204
```

Operations on UserEquipmentSpecification

List or find UserEquipmentSpecification objects

`GET /userEquipmentSpecification?fields=…`

Description

This operation list UserEquipmentSpecification entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a list of User Equipment Specifications. The given criteria is the @type (UserEquipmentSpecification) and hasNetworkInterface (true).

Request

```
GET
/userEquipmentSpecification?fields=id,href,hasNetworkInterface,isWearable,isPortable
,@type&hasNetworkInterface=true&isWearable=false
Content-Type: application/json
```

Response

```
200

[
    {
        "id": "11140",
        "href":
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11140",
        "description": "This is a first example of a user equipment specification
that we want for or binded UE.",
        "version": "1.0",
        "isPortable": false,
        "isWearable": false,
        "hasNetworkInterface": true,
        "vendor": "MyUEVendor",
        "model": "MyUEModel",
        "name": "MySpecificationName",
        "isBundle": false,
        "resourceSpecCharacteristic":[
            {
                "name": "MSISDN",
                "valueType": "text",
                "@type" : "CharacteristicSpecification"
            }
        ],
        "@type": "UserEquipmentSpecification",
```

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```
        "@schemaLocation":
"http://server:port/UserEquipmentSpecification.schema.json",
        "@baseType": "PhysicalResourceSpecification"
    },
    {
        "id": "11142",
        "href":
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11141",
        "description": "This is a second example of a user equipment specification
that we want for or binded UE.",
        "version": "1.0",
        "isPortable": true,
        "isWearable": false,
        "hasNetworkInterface": true,
        "vendor": "MyUEVendor",
        "model": "MyUEModel2",
        "name": "MySpecificationName2",
        "isBundle": false,
        "resourceSpecCharacteristic":[
            {
                "name": "MSISDN",
                "valueType": "text",
                "@type" : "CharacteristicSpecification"
            },
            {
                "name": "IMSI",
                "valueType": "text",
                "@type" : "CharacteristicSpecification"
            }
        ],
        "@type": "UserEquipmentSpecification",
        "@schemaLocation":
"http://server:port/UserEquipmentSpecification.schema.json",
        "@baseType": "PhysicalResourceSpecification"
    },
    {
        "id": "11143",
        "href":
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11143",
        "description": "This is an example of a user equipment specification that we
want for or binded UE.",
        "version": "1.0",
        "isPortable": false,
        "isWearable": false,
        "hasNetworkInterface": true,
        "vendor": "MyUEVendor",
        "model": "MyUEModel",
        "name": "MySpecificationName",
        "isBundle": false,
        "featureSpecification":[
            {
                "name": "privateOptimizedBindingFeature",
                "featureSpecRelationship" :[
                    {
                        "relationshipType" : "requires",
                        "name" : "privateOptimizedBindingFeature",
                        "@type" : "FeatureSpecificationRelationship"
                    }
                ],
                "featureSpecCharacteristic" : [
                    {
                        "name": "privateOptimizedBindingCloudAppHref",
                        "valueType": "text",
```


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```
                        "@type": "CharacteristicSpecification"
                    }
                ],
                "@type": "FeatureSpecification"
            }
        ],
        "resourceSpecCharacteristic":[
            {
                "name": "MSISDN",
                "valueType": "text",
                "@type": "CharacteristicSpecification"}
        ],
        "@type": "UserEquipmentSpecification",
        "@schemaLocation":
"http://server:port/UserEquipmentSpecification.schema.json",
        "@baseType": "PhysicalResourceSpecification"
    }
]
```

Retrieves a UserEquipmentSpecification by ID

GET /userEquipmentSpecification/{id}?fields=…

Description

This operation retrieves a UserEquipmentSpecification entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving an UserEquipmentSpecification.

Request

```
GET /userEquipmentSpecification/11140
Content-Type: application/json
```

Response

```
200

{
    "id": "11140",
    "href":
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11140",
    "version": "1.0",
    "isPortable": false,
    "isWearable": false,
    "hasNetworkInterface": true,
    "vendor": "MyUEVendor",
    "model": "MyUEModel",
    "name": "MySpecificationName",
    "resourceSpecCharacteristic":[
        {
```

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```
            "name": "MSISDN",
            "valueType": "text",
            "@type": "CharacteristicSpecification"
        }
    ],
    "@type": "UserEquipmentSpecification",
    "@schemaLocation": "http://server:port/UserEquipmentSpecification.schema.json",
    "@baseType": "PhysicalResourceSpecifcation"
}
```

Here's an example of a request for retrieving an UserEquipmentSpecification with BindingFeature.

Request

```
GET /userEquipmentSpecification/11143
Content-Type: application/json
```

Response

```
200

{
    "id": "11143",
    "href":
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11143",
    "version": "1.0",
    "isPortable": false,
    "isWearable": false,
    "hasNetworkInterface": true,
    "vendor": "MyUEVendor",
    "model": "MyUEModel",
    "name": "MySpecificationNameBinded",
    "featureSpecification":[
        {
            "name": "privateOptimizedBindingFeature",
            "featureSpecRelationship" :[
                {
                    "relationshipType" : "requires",
                    "name" : "privateOptimizedBindingFeature",
                    "@type" : "FeatureSpecificationRelationship"
                }
            ],
            "featureSpecCharacteristic" : [
                {
                    "name": "privateOptimizedBindingCloudAppHref",
                    "valueType": "text",
                    "@type": "CharacteristicSpecification"
                }
            ],
            "@type": "FeatureSpecification"
        }
    ],
    "resourceSpecCharacteristic":[
        {
            "name": "MSISDN",
            "valueType": "text",
```

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```
            "@type": "CharacteristicSpecification"
        }
    ],
    "@type": "UserEquipmentSpecification",
    "@schemaLocation": "http://server:port/UserEquipmentSpecification.schema.json",
    "@baseType": "PhysicalResourceSpecifcation"
}
```

Creates a UserEquipmentSpecification

`POST /userEquipmentSpecification?fields=…`

Description

This operation creates a UserEquipmentSpecification entity.

Mandatory Attributes

Mandatory Attributes	Rule
hasNetworkInterface	
isPortable	
isWearable	
@type	

Usage samples

Here's an example of a request and response for creating a new UserEquipmentSpecification

Request

```
POST /userEquipmentSpecification
Content-Type: application/json

{
    "description": "This is an example of a user equipment specification that we
want for or binded UE.",
    "version": "1.0",
    "isPortable": false,
    "isWearable": false,
    "hasNetworkInterface": true,
    "vendor": "MyUEVendor",
    "model": "MyUEModel",
    "name": "MySpecificationName",
    "isBundle": false,
    "resourceSpecCharacteristic":[
        {
            "name": "MSISDN",
            "valueType": "text",
            "@type": "CharacteristicSpecification"
        }
    ],
    "@type": "UserEquipmentSpecification",
    "@schemaLocation": "http://server:port/UserEquipmentSpecification.schema.json",
```

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```
    "@baseType": "PhysicalResourceSpecification"
}
```

Response

```
201

{
    "id": "11140",
    "href":
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11140",
    "description": "This is an example of a user equipment specification that we
want for or binded UE.",
    "version": "1.0",
    "isPortable": false,
    "isWearable": false,
    "hasNetworkInterface": true,
    "vendor": "MyUEVendor",
    "model": "MyUEModel",
    "name": "MySpecificationName",
    "isBundle": false,
    "resourceSpecCharacteristic":[
        {
            "name": "MSISDN",
            "valueType": "text",
            "@type": "CharacteristicSpecification"
        }
    ],
    "@type": "UserEquipmentSpecification",
    "@schemaLocation": "http://server:port/UserEquipmentSpecification.schema.json",
    "@baseType": "PhysicalResourceSpecification"
}
```

Updates partially a UserEquipmentSpecification

`PATCH /userEquipmentSpecification/{id}?fields=…`

Description

This operation allows partial updates of a UserEquipmentSpecification entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optiona. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
hasNetworkInterface	
href	
id	

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Non Patchable Attributes	Rule
isPortable	
isWearable	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
attachment	
category	
description	
externalIdentifier	
featureSpecification	
intentSpecification	
isBundle	
lastUpdate	
lifecycleStatus	
model	
name	
part	
relatedParty	
resourceSpecCharacteristic	
resourceSpecRelationship	
sku	
targetResourceSchema	
validFor	
vendor	
version	

Usage samples

Update a UserEquipmentSpecification using json

Request

```
PATCH /userEquipmentSpecification/{id}
Content-Type: application/json

{
```

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```
    "isBundle": false,
    "description": "my set of video cameras for surveillance",
    "@type": "UserEquipmentSpecification"
}
```

Response

```
200

{
    "id": "11140",
    "href":
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11140",
    "isBundle": false,
    "isWearable": false,
    "isPortable" : false,
    "hasNetworkInterface": true,
    "description": "my set of video cameras for surveillance",
    "resourceSpecCharacteristic":[
        {
            "name": "MSISDN",
            "valueType": "text",
            "@type": "CharacteristicSpecification"
        }
    ],
    "@type": "UserEquipmentSpecification",
    "@schemaLocation": "http://server:port/UserEquipmentSpecification.schema.json",
    "@baseType": "PhysicalResourceSpecifcation"
}
```

Update a UserEquipmentSpecification using json-patch and query parameters

Request

```
PATCH
/userEquipmentSpecification/{id}?fields=id,href,hasNetworkInterface,isWearable,isPor
table,@type
Content-Type: application/json-patch+json

[
    {   "op":"replace",
        "path": "/isBundle",
        "value": "false"
    },
    {   "op":"replace",
        "path" : "/description",
        "value" : "my set of video cameras for surveillance"
    }
]
```

Response

```
200

{
```

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```
    "id": "11140",
    "href":
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11140",
    "isWearable": false,
    "isPortable" : false,
    "hasNetworkInterface": true,
    "description": "my set of video cameras for surveillance",
    "resourceSpecCharacteristic":[
        {
            "name": "MSISDN",
            "valueType": "text",
            "@type": "CharacteristicSpecification"
        }
    ],
    "@type": "UserEquipmentSpecification"
}
```

Update a UserEquipmentSpecification using json-patch but no query parameters

Request

```
PATCH /userEquipmentSpecification/{id}
Content-Type: application/json-patch+json

[
    {   "op":"replace",
        "path": "/isBundle",
        "value": "false"
    },
    {   "op":"replace",
        "path" : "/description",
        "value" : "my set of video cameras for surveillance"
    }
]
```

Response

```
200

{
    "id": "11140",
    "href":
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11140",
    "isBundle": false,
    "isWearable": false,
    "isPortable" : false,
    "hasNetworkInterface": true,
    "description": "my set of video cameras for surveillance",
    "resourceSpecCharacteristic":[
        {
            "name": "MSISDN",
            "valueType": "text",
            "@type": "CharacteristicSpecification"
        }
    ],
    "@type": "UserEquipmentSpecification",
    "@schemaLocation": "http://server:port/UserEquipmentSpecification.schema.json",
```

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```
    "@baseType": "PhysicalResourceSpecifcation"
}
```

Update a UserEquipmentSpecification using json-patch-query

Request

```
PATCH /userEquipmentSpecification/{id}
Content-Type: application/json-patch-query+json

[
    {   "op":"replace",
        "path": "/isBundle?isBundle=true",
        "value": false
    },
    {   "op":"replace",
        "path" : "/description?isBundle=true",
        "value" : "my set of video cameras for surveillance"
    }
]
```

Response

```
200

{
    "id": "11140",
    "href":
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11140",
    "isBundle": false,
    "isWearable": false,
    "isPortable" : false,
    "hasNetworkInterface": true,
    "description": "my set of video cameras for surveillance",
    "resourceSpecCharacteristic":[
        {
            "name": "MSISDN",
            "valueType": "text",
            "@type": "CharacteristicSpecification"
        }
    ],
    "@type": "UserEquipmentSpecification",
    "@schemaLocation": "http://server:port/UserEquipmentSpecification.schema.json",
    "@baseType": "PhysicalResourceSpecifcation"
}
```

Update a UserEquipmentSpecification using merge-patch

Request

```
PATCH /userEquipmentSpecification/{id}
Content-Type: application/merge-patch+json

{
    "isBundle": false,
```

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```
    "description": "my set of video cameras for surveillance",
    "@type": "UserEquipmentSpecification"
}
```

Response

```
200

{
    "id": "11140",
    "href":
"http://server:port/privateOptimizedBinding/v5/userEquipmentSpecification/11140",
    "isBundle": false,
    "isWearable": false,
    "isPortable" : false,
    "hasNetworkInterface": true,
    "description": "my set of video cameras for surveillance",
    "resourceSpecCharacteristic":[
        {
            "name": "MSISDN",
            "valueType": "text",
            "@type": "CharacteristicSpecification"
        }
    ],
    "@type": "UserEquipmentSpecification",
    "@schemaLocation": "http://server:port/UserEquipmentSpecification.schema.json",
    "@baseType": "PhysicalResourceSpecifcation"
}
```

Deletes a UserEquipmentSpecification

`DELETE /userEquipmentSpecification/{id}`

Description

This operation deletes a UserEquipmentSpecification entity.

Usage samples

Here's an example for a request for deleteing specific UserEquipmentSpecification.

Request

```
DELETE /userEquipmentSpecification/11140
Content-Type: application/json
```

Response

```
204
```

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Operations on CloudApplicationSpecification

List or find CloudApplicationSpecification objects

`GET /cloudApplicationSpecification?fields=…`

Description

This operation list CloudApplicationSpecification entities. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving a list of User Cloud Application Specifications. The given criteria is the @type (CloudApplicationSpecification), isBundle and the version.

Request

```
GET
/cloudApplicationSpecification?fields=id,href,isBundle,version,@type&isBundle=true&v
ersion="1.0"
Content-Type: application/json
```

Response

```
200

[
    {
        "id": "11140",
        "href":
"http://server:port/privateOptimizedBinding/v5/cloudApplicationSpecification/11140",
        "description": "This is an example of a cloud application specification.",
        "version": "1.0",
        "isBundle": false,
        "name": "MyAppSpecificationName",
        "resourceSpecCharacteristic":[
            {
                "name": "cloudProvider",
                "valueType": "text",
                "@type" : "CharacteristicSpecification"
            },
            {
                "name": "cloudRegion",
                "valueType": "text",
                "@type" : "CharacteristicSpecification"
            },
            {
                "name": "cloudReference",
                "valueType": "text",
                "@type" : "CharacteristicSpecification"
            }
        ],
        "@type": "CloudApplicationSpecification",
        "@schemaLocation":
```

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```
"http://server:port/CloudApplicationSpecification.schema.json",
        "@baseType": "LogicalResourceSpecification"
    },
    {
        "id": "11141",
        "href":
"http://server:port/privateOptimizedBinding/v5/cloudApplicationSpecification/11141",
        "description": "This is an example of a cloud application specification.",
        "version": "1.0",
        "isBundle": false,
        "name": "MyGameSpecificationName",
        "resourceSpecCharacteristic":[
            {
                "name": "cloudProvider",
                "valueType": "text",
                "@type" : "CharacteristicSpecification"
            },
            {
                "name": "cloudRegion",
                "valueType": "text",
                "@type" : "CharacteristicSpecification"
            },
            {
                "name": "cloudReference",
                "valueType": "text",
                "@type" : "CharacteristicSpecification"
            },
            {
                "name": "myGameCharacteristic",
                "valueType": "text",
                "@type" : "CharacteristicSpecification"
            }
        ],
        "@type": "CloudApplicationSpecification",
        "@schemaLocation":
"http://server:port/CloudApplicationSpecification.schema.json",
        "@baseType": "LogicalResourceSpecification"
    }
]
```

Retrieves a CloudApplicationSpecification by ID

`GET /cloudApplicationSpecification/{id}?fields=…`

Description

This operation retrieves a CloudApplicationSpecification entity. Attribute selection is enabled for all first level attributes. Filtering may be available depending on the compliance level supported by an implementation.

Usage samples

Here's an example of a request for retrieving an CloudApplicationSpecification.

Request

```
GET /cloudApplicationSpecification/11140
```

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```
Content-Type: application/json
```

Response

```
200

{
    "id": "11140",
    "href":
"http://server:port/privateOptimizedBinding/v5/cloudApplicationSpecification/11140",
    "description": "This is an example of a cloud application specification.",
    "version": "1.0",
    "isBundle": true,
    "name": "MyAppSpecificationName",
    "resourceSpecCharacteristic":[
        {
            "name": "cloudProvider",
            "valueType": "text",
            "@type" : "CharacteristicSpecification"
        },
        {
            "name": "cloudRegion",
            "valueType": "text",
            "@type" : "CharacteristicSpecification"
        },
        {
            "name": "cloudReference",
            "valueType": "text",
            "@type" : "CharacteristicSpecification"
        }
    ],
    "@type": "CloudApplicationSpecification",
    "@schemaLocation":
"http://server:port/CloudApplicationSpecification.schema.json",
    "@baseType": "LogicalResourceSpecification"
}
```

Creates a CloudApplicationSpecification

`POST /cloudApplicationSpecification?fields=…`

Description

This operation creates a CloudApplicationSpecification entity.

Mandatory Attributes

Mandatory Attributes	Rule
@type	

Usage samples

Here's an example of a request and response for creating a new CloudApplicationSpecification

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Request

```
POST /cloudApplicationSpecification
Content-Type: application/json

{
    "description": "This is an example of a cloud application specification.",
    "version": "1.0",
    "name": "MyAppSpecificationName",
    "resourceSpecCharacteristic":[
        {
            "name": "cloudProvider",
            "valueType": "text",
            "@type" : "CharacteristicSpecification"
        },
        {
            "name": "cloudRegion",
            "valueType": "text",
            "@type" : "CharacteristicSpecification"
        },
        {
            "name": "cloudReference",
            "valueType": "text",
            "@type" : "CharacteristicSpecification"
        }
    ],
    "@type": "CloudApplicationSpecification",
    "@schemaLocation":
"http://server:port/CloudApplicationSpecification.schema.json",
    "@baseType": "LogicalResourceSpecification"
}
```

Response

```
201

{
    "id": "11140",
    "href":
"http://server:port/privateOptimizedBinding/v5/cloudApplicationSpecification/11140",
    "description": "This is an example of a cloud application specification.",
    "version": "1.0",
    "name": "MyAppSpecificationName",
    "resourceSpecCharacteristic":[
        {
            "name": "cloudProvider",
            "valueType": "text",
            "@type" : "CharacteristicSpecification"
        },
        {
            "name": "cloudRegion",
            "valueType": "text",
            "@type" : "CharacteristicSpecification"
        },
        {
            "name": "cloudReference",
            "valueType": "text",
            "@type" : "CharacteristicSpecification"
        }
```

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```
    ],
    "@type": "CloudApplicationSpecification",
    "@schemaLocation":
"http://server:port/CloudApplicationSpecification.schema.json",
    "@baseType": "LogicalResourceSpecification"
}
```

Updates partially a CloudApplicationSpecification

`PATCH /cloudApplicationSpecification/{id}?fields=…`

Description

This operation allows partial updates of a CloudApplicationSpecification entity. Support of json/merge (https://tools.ietf.org/html/rfc7386) is mandatory, support of json/patch (http://tools.ietf.org/html/rfc5789) is optiona. Note: If the update operation yields to the creation of sub-resources or relationships, the same rules concerning mandatory sub-resource attributes and default value settings in the POST operation applies to the PATCH operation. Hence these tables are not repeated here.

Patchable and Non Patchable Attributes

Non Patchable Attributes	Rule
href	
id	
version	
@baseType	@baseType is immutable
@schemaLocation	@schemaLocation is immutable
@type	@type is immutable

Patchable Attributes	Rule
attachment	
category	
description	
externalIdentifier	
featureSpecification	
intentSpecification	
isBundle	
lastUpdate	
lifecycleStatus	
name	
relatedParty	
resourceSpecCharacteristic	

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Patchable Attributes	Rule
resourceSpecRelationship	
targetResourceSchema	
validFor	

Usage samples

Update a CloudApplicationSpecification using json

Request

```
PATCH /cloudApplicationSpecification/{id}
Content-Type: application/json

{
    "isBundle": false,
    "description": "my new cloud application game",
    "@type": "CloudApplicationSpecification"
}
```

Response

```
200

{
    "id": "11140",
    "href":
"http://server:port/privateOptimizedBinding/v5/cloudApplicationSpecification/11140",
    "description": "my new cloud application game",
    "isBundle": false,
    "name": "MyAppSpecificationName",
    "resourceSpecCharacteristic":[
        {
            "name": "cloudProvider",
            "valueType": "text",
            "@type" : "CharacteristicSpecification"
        },
        {
            "name": "cloudRegion",
            "valueType": "text",
            "@type" : "CharacteristicSpecification"
        },
        {
            "name": "cloudReference",
            "valueType": "text",
            "@type" : "CharacteristicSpecification"
        }
    ],
    "@type": "CloudApplicationSpecification",
    "@schemaLocation":
"http://server:port/CloudApplicationSpecification.schema.json",
    "@baseType": "LogicalResourceSpecification"
}
```

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Update a CloudApplicationSpecification using json-patch and query parameters

Request

```
PATCH /cloudApplicationSpecification/{id}?fields=id,href,description,isBundle,@type
Content-Type: application/json-patch+json

[
    {   "op":"replace",
        "path": "/isBundle",
        "value": "false"
    },
    {   "op":"replace",
        "path" : "/description",
        "value" : "my new cloud application game"
    }
]
```

Response

```
200

{
    "id": "11140",
    "href":
"http://server:port/privateOptimizedBinding/v5/cloudApplicationSpecification/11140",
    "description": "my new cloud application game",
    "name": "MyAppSpecificationName",
    "isBundle": false,
    "resourceSpecCharacteristic":[
        {
            "name": "cloudProvider",
            "valueType": "text",
            "@type" : "CharacteristicSpecification"
        },
        {
            "name": "cloudRegion",
            "valueType": "text",
            "@type" : "CharacteristicSpecification"
        },
        {
            "name": "cloudReference",
            "valueType": "text",
            "@type" : "CharacteristicSpecification"
        }
    ],
    "@type": "CloudApplicationSpecification",
    "@schemaLocation":
"http://server:port/CloudApplicationSpecification.schema.json",
    "@baseType": "LogicalResourceSpecification"
}
```

Update a CloudApplicationSpecification using json-patch but no query parameters

Request

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```
PATCH /cloudApplicationSpecification/{id}
Content-Type: application/json-patch+json

[
    {   "op":"replace",
        "path": "/isBundle",
        "value": "false"
    },
    {   "op":"replace",
        "path" : "/description",
        "value" : "my new cloud application game"
    }
]
```

Response

```
200

{
    "id": "11140",
    "href":
"http://server:port/privateOptimizedBinding/v5/cloudApplicationSpecification/11140",
    "description": "my new cloud application game",
    "isBundle": false,
    "name": "MyAppSpecificationName",
    "resourceSpecCharacteristic":[
        {
            "name": "cloudProvider",
            "valueType": "text",
            "@type" : "CharacteristicSpecification"
        },
        {
            "name": "cloudRegion",
            "valueType": "text",
            "@type" : "CharacteristicSpecification"
        },
        {
            "name": "cloudReference",
            "valueType": "text",
            "@type" : "CharacteristicSpecification"
        }
    ],
    "@type": "CloudApplicationSpecification",
    "@schemaLocation":
"http://server:port/CloudApplicationSpecification.schema.json",
    "@baseType": "LogicalResourceSpecification"
}
```

Update a CloudApplicationSpecification using json-patch-query

Request

```
PATCH /cloudApplicationSpecification/{id}
Content-Type: application/json-patch-query+json

[
```

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```
    {   "op":"replace",
        "path": "/isBundle?isBundle=true",
        "value": false
    },
    {   "op":"replace",
        "path" : "/description?isBundle=true",
        "value" : "my new cloud application game"
    }
]
```

Response

```
200

{
    "id": "11140",
    "href":
"http://server:port/privateOptimizedBinding/v5/cloudApplicationSpecification/11140",
    "description": "my new cloud application game",
    "isBundle": false,
    "name": "MyAppSpecificationName",
    "resourceSpecCharacteristic":[
        {
            "name": "cloudProvider",
            "valueType": "text",
            "@type" : "CharacteristicSpecification"
        },
        {
            "name": "cloudRegion",
            "valueType": "text",
            "@type" : "CharacteristicSpecification"
        },
        {
            "name": "cloudReference",
            "valueType": "text",
            "@type" : "CharacteristicSpecification"
        }
    ],
    "@type": "CloudApplicationSpecification",
    "@schemaLocation":
"http://server:port/CloudApplicationSpecification.schema.json",
    "@baseType": "LogicalResourceSpecification"
}
```

Update a CloudApplicationSpecification using merge-patch

Request

```
PATCH /cloudApplicationSpecification/{id}
Content-Type: application/merge-patch+json

{
    "isBundle": false,
    "description": "my new cloud application game",
    "@type": "CloudApplicationSpecification"
}
```

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Response

```
200

{
    "id": "11140",
    "href":
"http://server:port/privateOptimizedBinding/v5/cloudApplicationSpecification/11140",
    "description": "my new cloud application game",
    "isBundle": false,
    "name": "MyAppSpecificationName",
    "resourceSpecCharacteristic":[
        {
            "name": "cloudProvider",
            "valueType": "text",
            "@type" : "CharacteristicSpecification"
        },
        {
            "name": "cloudRegion",
            "valueType": "text",
            "@type" : "CharacteristicSpecification"
        },
        {
            "name": "cloudReference",
            "valueType": "text",
            "@type" : "CharacteristicSpecification"
        }
    ],
    "@type": "CloudApplicationSpecification",
    "@schemaLocation":
"http://server:port/CloudApplicationSpecification.schema.json",
    "@baseType": "LogicalResourceSpecification"
}
```

Deletes a CloudApplicationSpecification

`DELETE /cloudApplicationSpecification/{id}`

Description

This operation deletes a CloudApplicationSpecification entity.

Usage samples

Here's an example for a request for deleteing specific CloudApplicationSpecification.

Request

```
DELETE /cloudApplicationSpecification/11140
Content-Type: application/json
```

Response

```
204
```

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API NOTIFICATIONS

For every single of operation on the entities use the following templates and provide sample REST notification POST calls.

It is assumed that the Pub/Sub uses the Register and UnRegister mechanisms described in the REST Guidelines reproduced below.

Register listener

POST /hub

Description

Sets the communication endpoint address the service instance must use to deliver information about its health state, execution state, failures and metrics. Subsequent POST calls will be rejected by the service if it does not support multiple listeners. In this case DELETE /api/hub/{id} must be called before an endpoint can be created again.

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 409 if request is not successful.

Usage Samples

Here's an example of a request for registering a listener.

Request

```
POST /api/hub

Accept: application/json

{
    "callback": "http://in.listener.com"
}
```

Response

```
201

Content-Type: application/json

Location: /api/hub/42

{
    "id":"42",
    "callback": "http://in.listener.com",
    "query":""
}
```

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Unregister listener

DELETE /hub/{id}

Description

Clears the communication endpoint address that was set by creating the Hub..

Behavior

Returns HTTP/1.1 status code 204 if the request was successful.

Returns HTTP/1.1 status code 404 if the resource is not found.

Usage Samples

Here's an example of a request for un-registering a listener.

Request

```
DELETE /api/hub/42

Accept: application/json
```

Response

```
204
```

Publish Event to listener

POST /client/listener

Description

Clears the communication endpoint address that was set by creating the Hub.

Provides to a registered listener the description of the event that was raised. The /client/listener url is the callback url passed when registering the listener.

Behavior

Returns HTTP/1.1 status code 201 if the service is able to set the configuration.

Usage Samples

Here's an example of a notification received by the listener. In this example "EVENT TYPE" should be replaced by one of the notification types supported by this API (see Notification resources Models section) and EVENT BODY refers to the data structure of the given notification type.

Request

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```
POST /client/listener

Accept: application/json

{
    "event": {
        EVENT BODY
    },
    "eventType": "EVENT_TYPE"
}
```

Response

```
201
```

For detailed examples on the general TM Forum notification mechanism, see the TMF REST Design Guidelines.

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Acknowledgements

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